Unit / Module Description:	Zynq UltraScale MPSoC + 9-axis IMU + 4x FPC
Unit / Module Number:	VCS3
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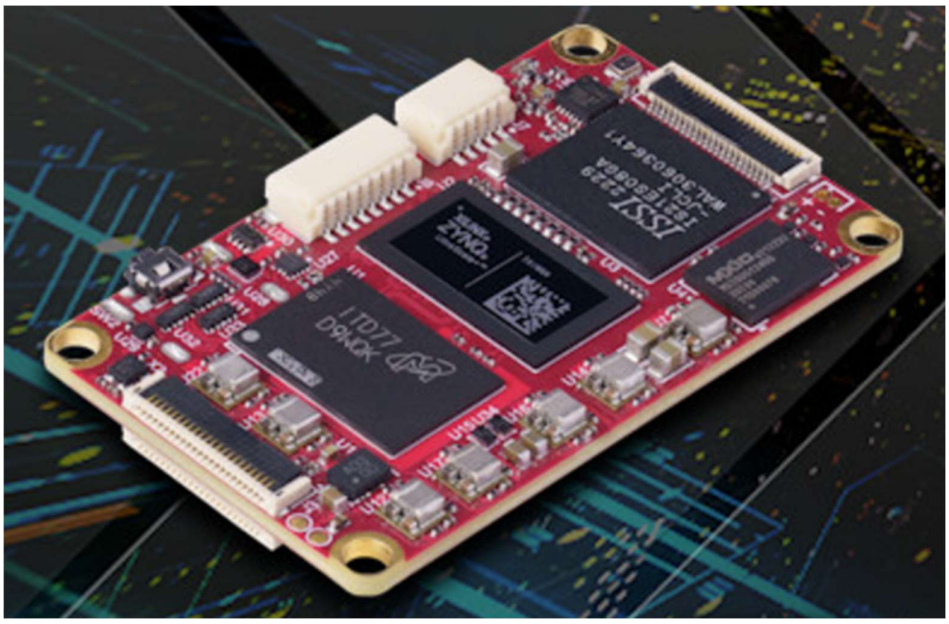
Issue Date:	20/01/2026
Original Author:	Paul Dorrell

VCS3 Application Starter's Guide



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Revision History

Issue	Changes Made	Date	Initials
1.0	First issue.	18/03/2025	PD
1.1	Added appendix (linked on p6, p39), added note on p38, corrected Fig. 48 to dual parallel flash type, added Fig. 50 and 51 on p42.	15/07/2025	NS
2.0	Updated for AMD/Xilinx 2025.2 tools	20/01/2026	PD

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1 Introduction

This document is a guide for those who own a VCS-3 and would like to develop applications to run on the Zynq MPSoC, either the PS (processor side) and/or the PL (programmable logic). The following are the basic steps to create simple projects using various AMD/Xilinx tools.

The hardware development kit is comprehensively described in the companion document "VCS3 Development Kit Getting Started Guide". The user is strongly advised to read this, as it contains details of the JTAG debug setup for application development.

2 Developing for VCS3 using Vivado and Vitis

The VCS3 has been used in Vivado versions 2022.2. 2023.2, 2024.2 and now 2025.2. It is recommended to use the latest version.

Sundance Multiprocessor Technology LTD provides documentation and resources on GitHub:

https://github.com/SundanceMultiprocessorTechnology/VCS-3.git

[see *section 5.1* for help on cloning this repository]

VCS3 board files are part of this repository, but can be downloaded separately from

https://github.com/SundanceMultiprocessorTechnology/VCS-3/tree/main/board_files

In the past, board files were typically copied into the Vivado installation path. However, this makes them specific to that particular Vivado version. An easier and better way is to copy them wherever you like and modify the vivado_init.tcl script to find them. On a typical Linux installation, this script can be found at ~/.Xilinx/Vivado.

Here is an example where the home directory is "/home/dev22" and the VCS3 board files have been copied to "/home/dev22/work/sundance/VCS3/board_files/".

```
set boardrepo [get_param board.repoPaths]
set sundancecustomRepo "/home/dev22/work/sundance/VCS3/board_files/"
set trenzCustomRepo "/home/dev22/work/trenz/te0950-rd/board_files/"
set digilentCustomRepo "/home/dev22/work/digilent/vivado-boards/new/board_files/"


if {[string first "My_boards_repo" [get_param board.repoPaths]] == -1} {
    set      boardrepos      "${boardrepo}      ${sundanceCustomRepo}      ${trenzCustomRepo}
${digilentCustomRepo}"
        set_param board.repoPaths $boardrepos
}
```

Alternatively, you can set Vivado to find the board files using the Tools Settings as shown in this example.

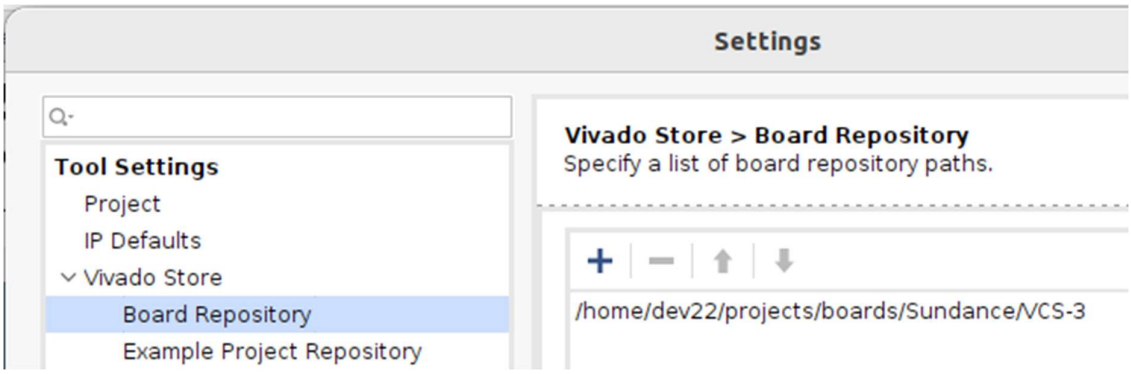


Figure 1 – Setting path to board files

Having Vivado open, create a new project, either on "Create Project", or File→Project→New:

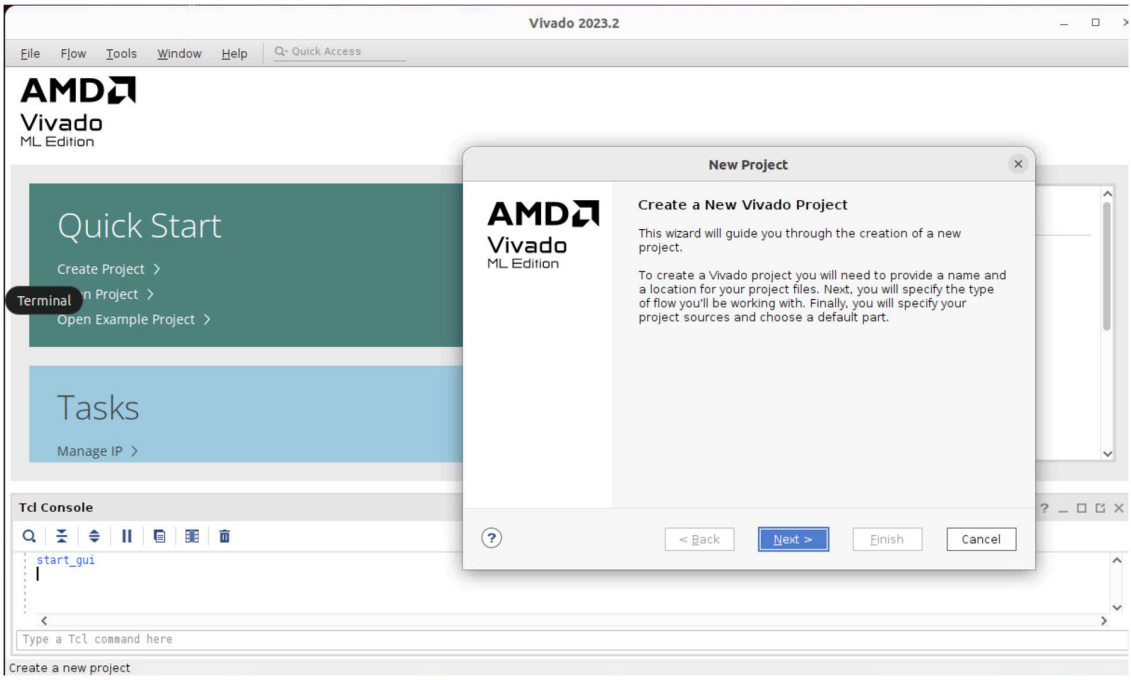


Figure 2 - Creating a new project

Click "Next" and select the path of the project.

Choose the RTL project and mark the square "Do not specify sources at this time" in case the user doesn't require importing any source, and it's a blank project.

Then, when Vivado asks for a device part, select "Boards" and choose the corresponding board files.

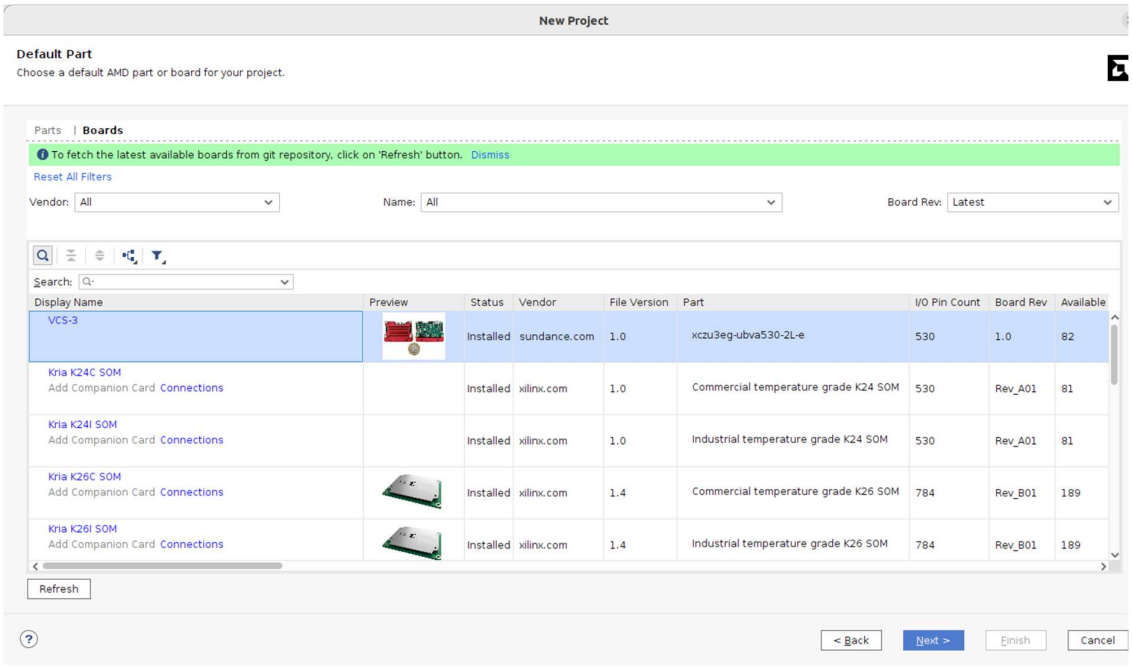


Figure 3 – Selecting board file

Once the project is opened, the user can see the information about the board and the part used:

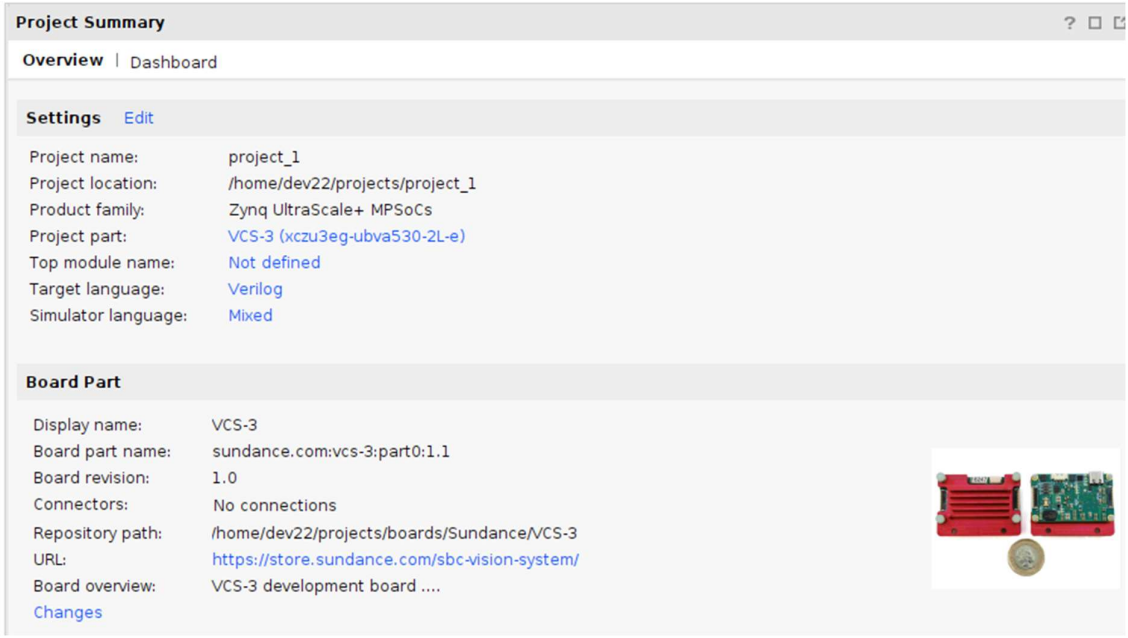


Figure 4 - Project information in Vivado

As soon as the user creates a new block design in IP Integrator, there will be some interfaces available, already set up and ready to use:

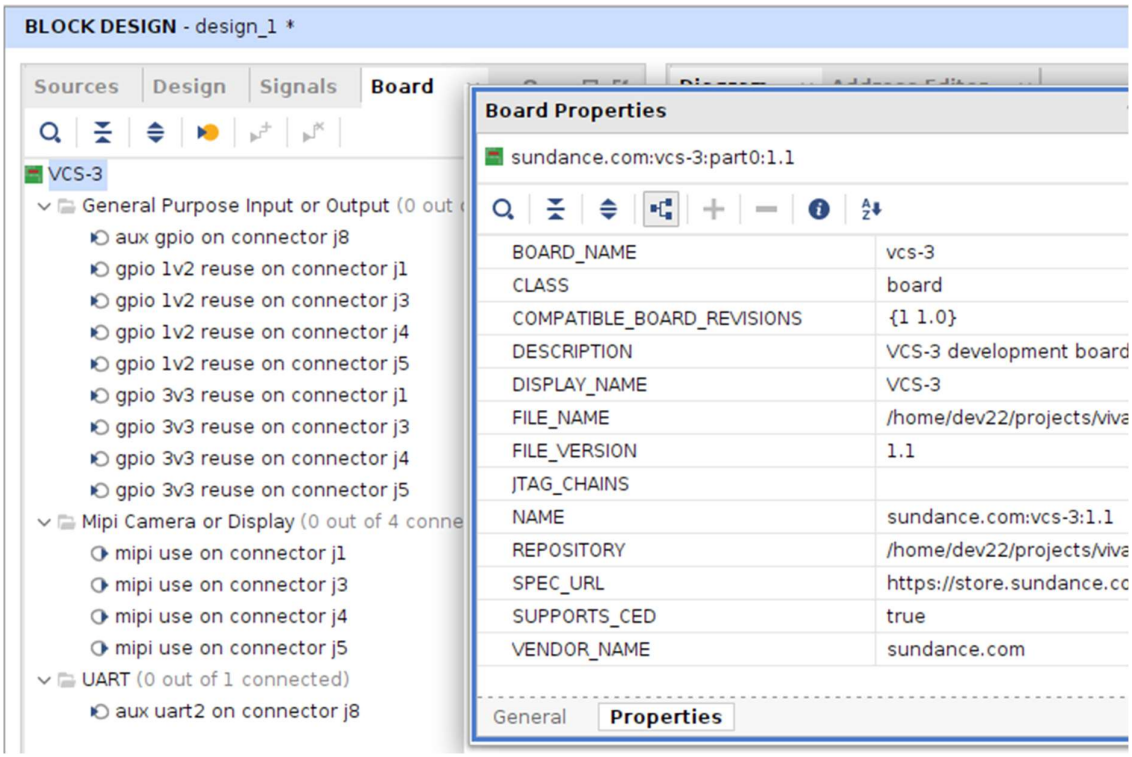


Figure 5 - Board interfaces in IP Integrator

The board interfaces available are those connected to the PL side of the Zynq. Other board interfaces are accessible through the Zynq PS MIO pins as described in the Hardware section.

3 Hardware

In addition to the "VCS3 Development Kit Getting Started Guide", the following sections show the raw pinout from the FPGA to peripherals or board connectors. However, board files are provided for easier interfacing with off-board peripherals such as cameras or displays. The board files also offer an easy way to repurpose some board connectors as "general purpose" io.

3.1 How can I setup the Zynq PS interfaces?

The processor side interfaces are easily setup by ensuring that "Apply Board Preset" is selected when adding "Zynq UltraScale+ MPSoc" IP core to your design and running "Block Automation":

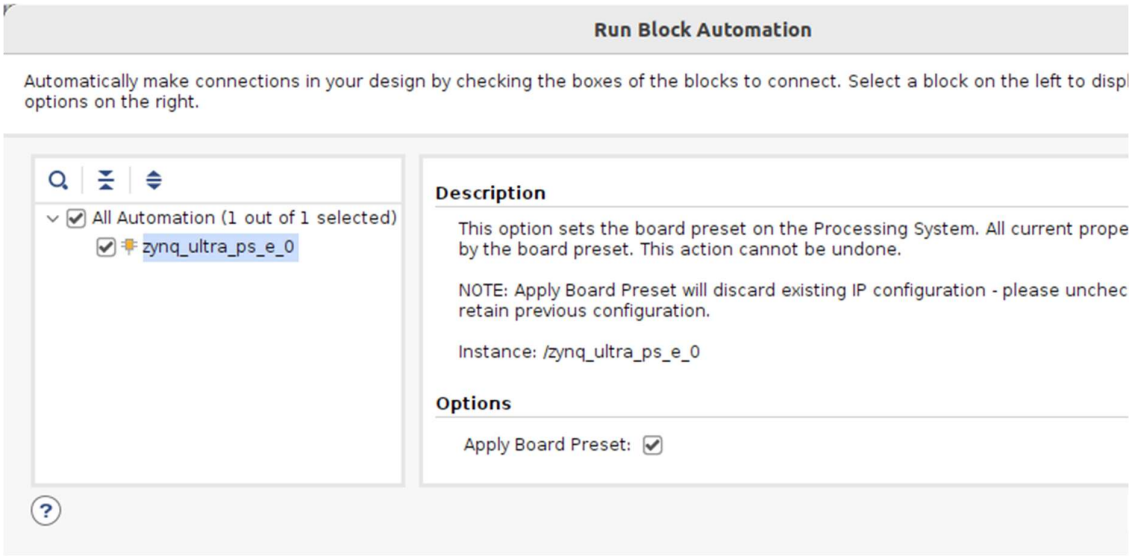


Figure 6 – Running Block Automation in IP Integrator

The resulting configuration can be viewed by recustomizing this block and cancelling when finished.

The preset configures PS interfaces for

- Clocks into programmable logic (PL) @ 100MHz, 200MHz, and 25MHz
- Quad SPI Flash
- SD (eMMC)
- CAN
- I2C
- SPI x2
- UART x2
- GPIO (x21 to PS pins, x1 into PL)
- Timer/counter x3
- USB (including USB 3.0)
- LPDDR4 memory controller

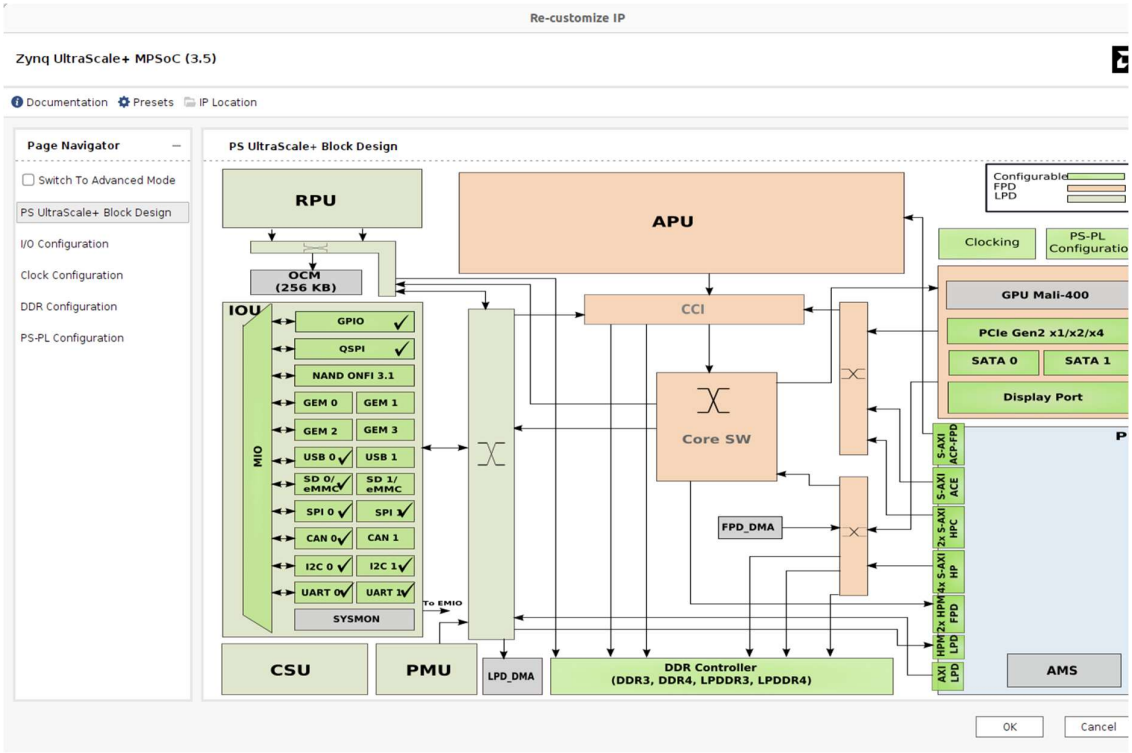


Figure 7 – Zynq configuration after Block Automation

3.2 How can I interface to the IMU?

The onboard 9-axis Inertial Measurement Unit (DS-000189-ICM-20948-v1.3.pdf) is connected using the following FPGA pins:

IMU device pin name	IMU device pin	board net name	connection type	FPGA pin	note
/CS	U5:22	IMU_SS0	direct connection	A12	PS_MIO67_502
SCL/SCLK	U5:23	IMU_SCLK	direct connection	B15	PS_MIO64_502
SDA/SDI	U5:24	IMU_MOSI	direct connection	A14	PS_MIO69_502
SDO/AD0	U5:9	IMU_MISO	direct connection	A13	PS_MIO68_502
FSYNC	U5:11	IMU_FSYNC	direct connection	B19	PS_MIO70_502
INT1	U5:12	IMU_INT	direct connection	A15	PS_MIO71_502

This peripheral is selected automatically when “Zynq UltraScale+ MPSoC” IP is added to a block design and the board preset is applied. It can then be accessed by the PS software using SPI 0.

3.3 How can I interface with external cameras or displays?

Four identical connectors (J1, J3, J4, and J5) can be used for any combination of external cameras and displays. These use the MIPI (Mobile Industry Processor Interface) standard, CSI (“camera serial interface”) or DSI (“display serial interface”). The following four connection tables are for CSI with the data lanes as inputs from the camera. For DSI, the connections are similar except that the data lanes are outputs to the display.

For each connector, data lanes are matched in length to within 0.5mm.

SPI control to each connector is done via an I2C to SMBus switch (pca9546a.pdf)

3.3.1 j1: camera_a interface

MIPI net name	connector pin	board net name	connection type	FPGA pin	note
CSI_CLK0_P	J1:9	CAM_A_CK_P	direct connection	N6	
CSI_CLK0_N	J1:8	CAM_A_CK_N	direct connection	P6	
CSI_DATA0_P	J1:3	CAM_A_D0_P	direct connection	N4	
CSI_DATA0_N	J1:2	CAM_A_D0_N	direct connection	P4	
CSI_DATA1_P	J1:6	CAM_A_D1_P	direct connection	M7	
CSI_DATA1_N	J1:5	CAM_A_D1_N	direct connection	N7	
CSI_DATA2_P	J1:12	CAM_A_D2_P	direct connection	T6	
CSI_DATA2_N	J1:11	CAM_A_D2_N	direct connection	U5	
CSI_DATA3_P	J1:15	CAM_A_D3_P	direct connection	T4	
CSI_DATA3_N	J1:14	CAM_A_D3_N	direct connection	U4	
	J1:17	CAM_A_EN	direct connection	K15	Under control of PS side of Zynq
	J1:18	CAM_A_CLK_28	via resistive divider	D10	CAM_A_CLK
	J1:20	CAM_A_SCL	indirect connection via I2C mux U6	J11	CAM_SCL
	J1:21	CAM_A_SDA	indirect connection via I2C mux U6	M10	CAM_SCD

3.3.2 j3: camera_b interface

MIPI net name	connector pin	board net name	connection type	FPGA pin	note
CSI_CLK0_P	J3:9	CAM_B_CK_P	direct connection	T2	
CSI_CLK0_N	J3:8	CAM_B_CK_N	direct connection	U2	
CSI_DATA0_P	J3:3	CAM_B_D0_P	direct connection	N2	
CSI_DATA0_N	J3:2	CAM_B_D0_N	direct connection	N1	
CSI_DATA1_P	J3:6	CAM_B_D1_P	direct connection	R1	
CSI_DATA1_N	J3:5	CAM_B_D1_N	direct connection	T1	
CSI_DATA2_P	J3:12	CAM_B_D2_P	direct connection	V4	
CSI_DATA2_N	J3:11	CAM_B_D2_N	direct connection	V3	
CSI_DATA3_P	J3:15	CAM_B_D3_P	direct connection	K7	
CSI_DATA3_N	J3:14	CAM_B_D3_N	direct connection	K6	
	J3:17	CAM_B_EN	direct connection	M12	Under control of PS side of Zynq
	J3:18	CAM_B_CLK_28	via resistive divider	D9	CAM_B_CLK
	J3:20	CAM_B_SCL	indirect connection via I2C mux U6	J11	CAM_SCL
	J3:21	CAM_B_SDA	indirect connection via I2C mux U6	M10	CAM_SCD

3.3.3 j4: camera_c interface

MIPI net name	connector pin	board net name	connection type	FPGA pin	note
CSI_CLK0_P	J4:9	CAM_C_CK_P	direct connection	K2	
CSI_CLK0_N	J4:8	CAM_C_CK_N	direct connection	L1	
CSI_DATA0_P	J4:3	CAM_C_D0_P	direct connection	L2	
CSI_DATA0_N	J4:2	CAM_C_D0_N	direct connection	M1	
CSI_DATA1_P	J4:6	CAM_C_D1_P	direct connection	G7	
CSI_DATA1_N	J4:5	CAM_C_D1_N	direct connection	H7	
CSI_DATA2_P	J4:12	CAM_C_D2_P	direct connection	J1	
CSI_DATA2_N	J4:11	CAM_C_D2_N	direct connection	K1	
CSI_DATA3_P	J4:15	CAM_C_D3_P	direct connection	H8	
CSI_DATA3_N	J4:14	CAM_C_D3_N	direct connection	J7	
	J4:17	CAM_C_EN	direct connection	N9	Under control of PS side of Zynq
	J4:18	CAM_C_CLK_28	via resistive divider	D13	CAM_C_CLK
	J4:20	CAM_C_SCL	indirect connection via I2C mux U6	J11	CAM_SCL
	J4:21	CAM_C_SDA	indirect connection via I2C mux U6	M10	CAM_SCD

3.3.4 j5: camera_d interface

MIPI net name	connector pin	board net name	connection type	FPGA pin	note
CSI_CLK0_P	J5:9	CAM_D_CK_P	direct connection	H2	
CSI_CLK0_N	J5:8	CAM_D_CK_N	direct connection	H1	
CSI_DATA0_P	J5:3	CAM_D_D0_P	direct connection	G2	
CSI_DATA0_N	J5:2	CAM_D_D0_N	direct connection	G1	
CSI_DATA1_P	J5:6	CAM_D_D1_P	direct connection	F7	
CSI_DATA1_N	J5:5	CAM_D_D1_N	direct connection	G6	
CSI_DATA2_P	J5:12	CAM_D_D2_P	direct connection	E1	
CSI_DATA2_N	J5:11	CAM_D_D2_N	direct connection	F1	
CSI_DATA3_P	J5:15	CAM_D_D3_P	direct connection	C1	
CSI_DATA3_N	J5:14	CAM_D_D3_N	direct connection	D1	
	J5:17	CAM_D_EN	direct connection	K12	Under control of PS side of Zynq
	J5:18	CAM_D_CLK_28	via resistive divider	C13	CAM_D_CLK
	J5:20	CAM_D_SCL	indirect connection via I2C mux U6	J11	CAM_SCL
	J5:21	CAM_D_SDA	indirect connection via I2C mux U6	M10	CAM_SCD

3.3.5 Using board file for mipi camera serial interface on j1, j3, j4, and j5

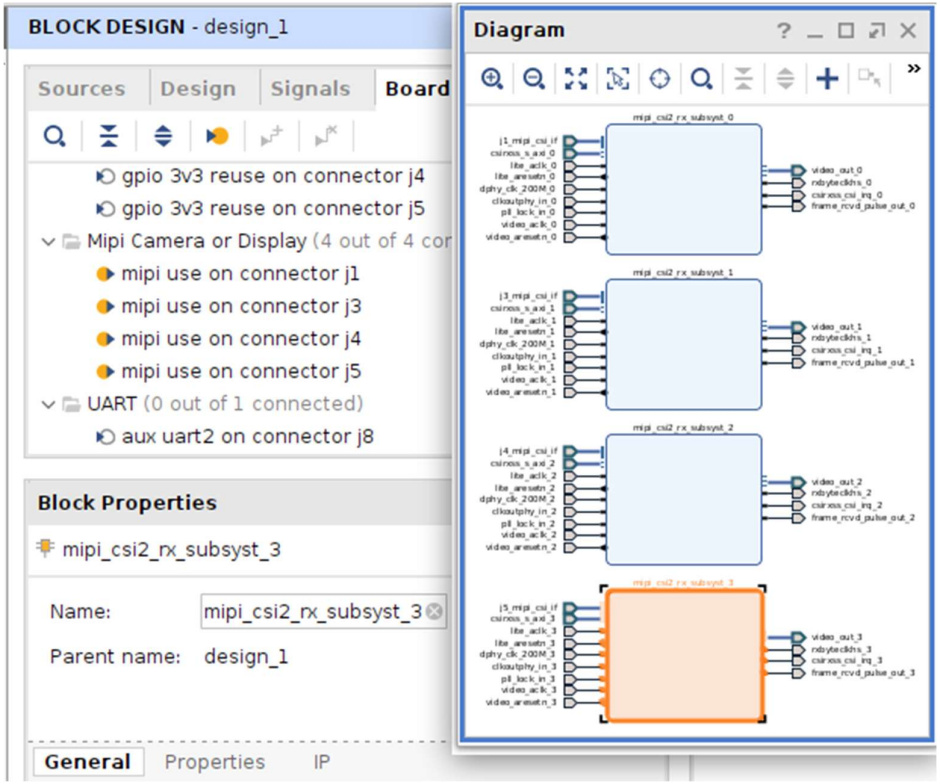


Figure 8 – board file mipi camera interfaces

3.3.6 Using board file for mipi display serial interface on j1 and j3

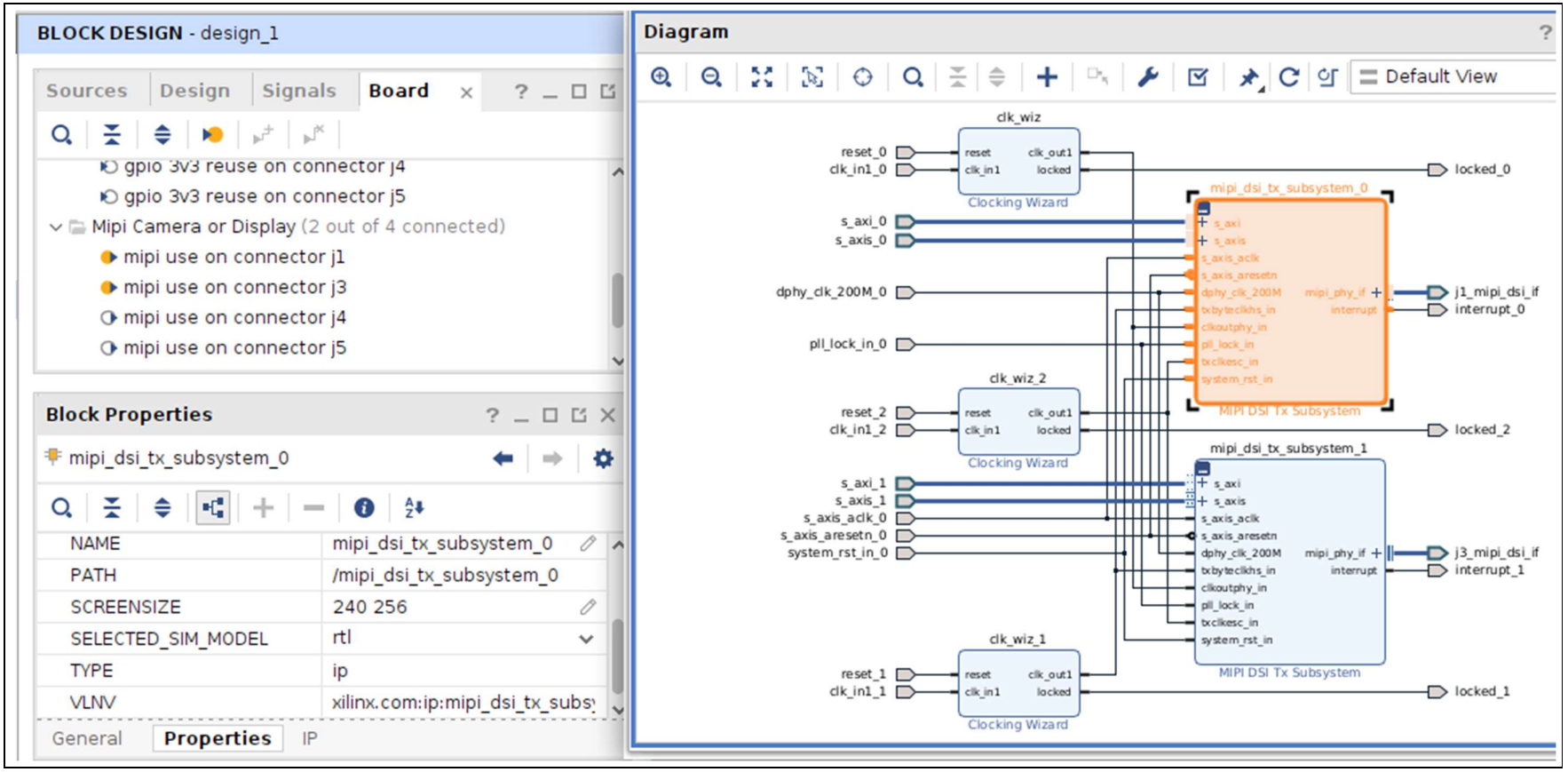


Figure 9 – board file mipi display interfaces

3.4 How can I repurpose the board connectors for GPIO?

Any of the four identical connectors (J1, J3, J4, and J5) can be repurposed for “General Purpose IO” using the Board tab. The image below shows a complete repurpose of all four. Note that one pin in each connector is at 3v3 logical standard and handled by the GPIO2 interface of each respective “AXI GPIO”.

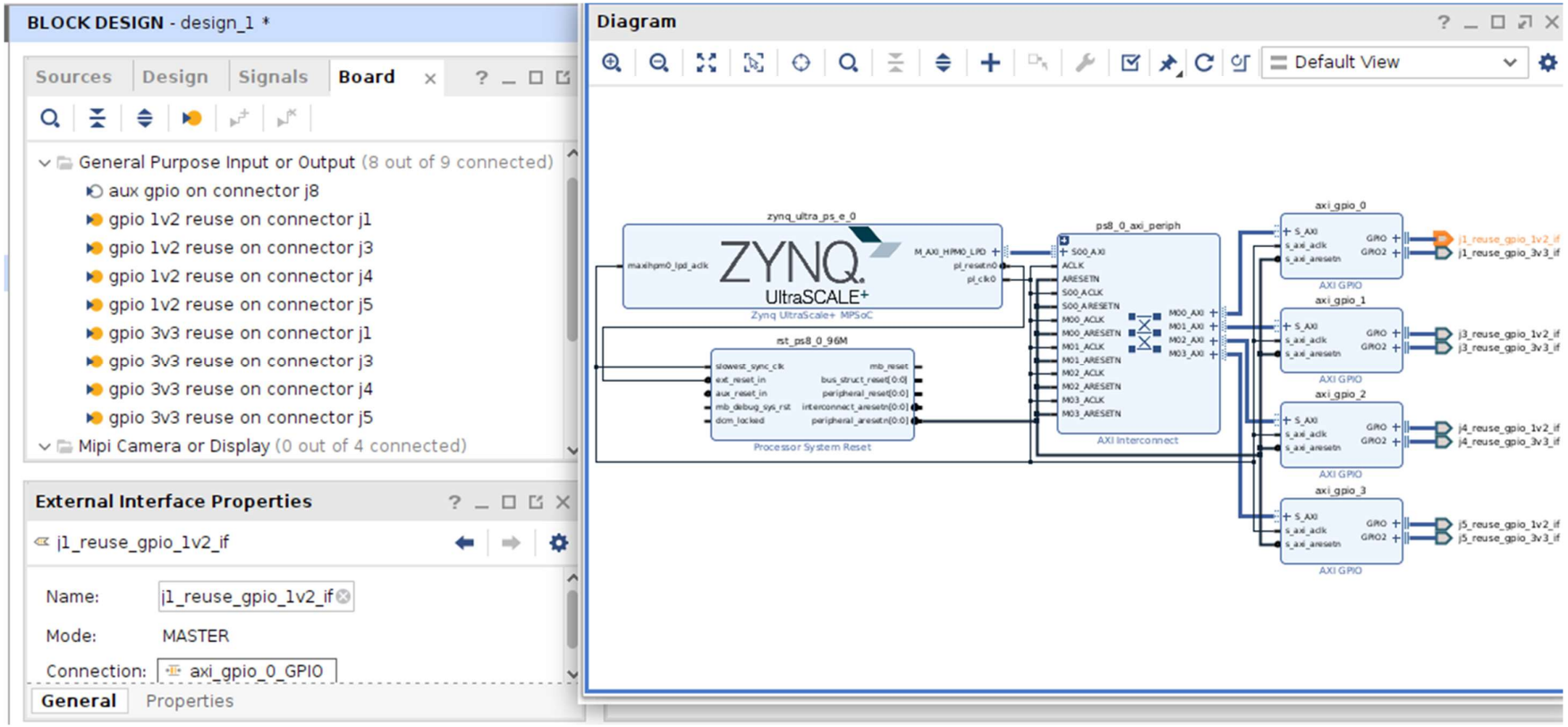


Figure 10 – board file repurposed gpio interfaces

3.5 What other interfaces are provided?

3.5.1 I2C for monitoring voltage supplies (U7, U13) etc

Device pin	board net name	connection type	FPGA pin	PS pin
U7:4 U13:4 U27:8 U21: G7 U11:9	SCL	direct connection	A18	PS_MIO74_502 Pulled up to 1V8 using 10k ohms
U7:1 U13:1 U27:7 U21: H8 U11:8	SDA	direct connection	A16	PS_MIO75_502 Pulled up to 1V8 using 10k ohms

This peripheral is selected automatically when “Zynq UltraScale+ MPSoC” IP is added to a block design and the board preset is applied. It may be accessed from PS software as I2C 0.

3.5.2 I2C to SPI camera/display control (U6)

Device pin	board net name	connection type	FPGA pin	PS pin
U6:14	CAM_SCL	direct connection	J11	PS_MIO28_501 Pulled up to 3V3 using 4k7 ohms
U6:15	CAM_SDA	direct connection	M10	PS_MIO29_501 Pulled up to 3V3 using 4k7 ohms

This peripheral is selected automatically when “Zynq UltraScale+ MPSoC” IP is added to a block design and the board preset is applied. It may be accessed from the PS software as I2C 1.

3.5.3 J7: can bus

J7

connector pin	connection type	board net name	FPGA pin	PS pin
J7:4	indirect connection	CAN_RXD	K10	PS_MIO51_501
J7:5	indirect connection	CAN_TXD	J10	PS_MIO50_501
		CAN_SCLK	M13	PS_MIO44_501
		CAN_MOSI	M14	PS_MIO49_501
		CAN_MISO	N12	PS_MIO48_501
		CAN_SS0	N13	PS_MIO47_501

This peripheral is selected automatically when “Zynq UltraScale+ MPSoC” IP is added to a block design and the board preset is applied. It may be accessed from PS software as SPI 1.

3.5.4 J9: uart0

J9

connector pin	board net name	connection type	FPGA pin	PS pin
J9:2	UART0_RX	direct connection	K14	PS_MIO30_501
J9:3	UART0_TX	direct connection	J12	PS_MIO31_501

This peripheral is selected automatically when “Zynq UltraScale+ MPSoC” IP is added to a block design and the board preset is applied. It may be accessed from the PS software as UART 0.

3.5.5 J7: uart1

J7

connector pin	board net name	connection type	FPGA pin	PS pin
J7:1	UART1_RX	direct connection	M9	PS_MIO32_501
J7:2	UART1_TX	direct connection	H14	PS_MIO33_501

This peripheral is selected automatically when “Zynq UltraScale+ MPSoC” IP is added to a block design and the board preset is applied. It may be accessed from PS software as UART 1.

3.5.6 J8: uart2

J8

connector pin	board net name	connection type	FPGA pin	note
J8:2	AUX_UART2_RX	direct connection	B10	
J8:3	AUX_UART2_TX	direct connection	C10	

This peripheral is selected when the “aux uart2 on connector j8” is selected from the Board tab or when “AXI Uartlite” IP is added to a block design.

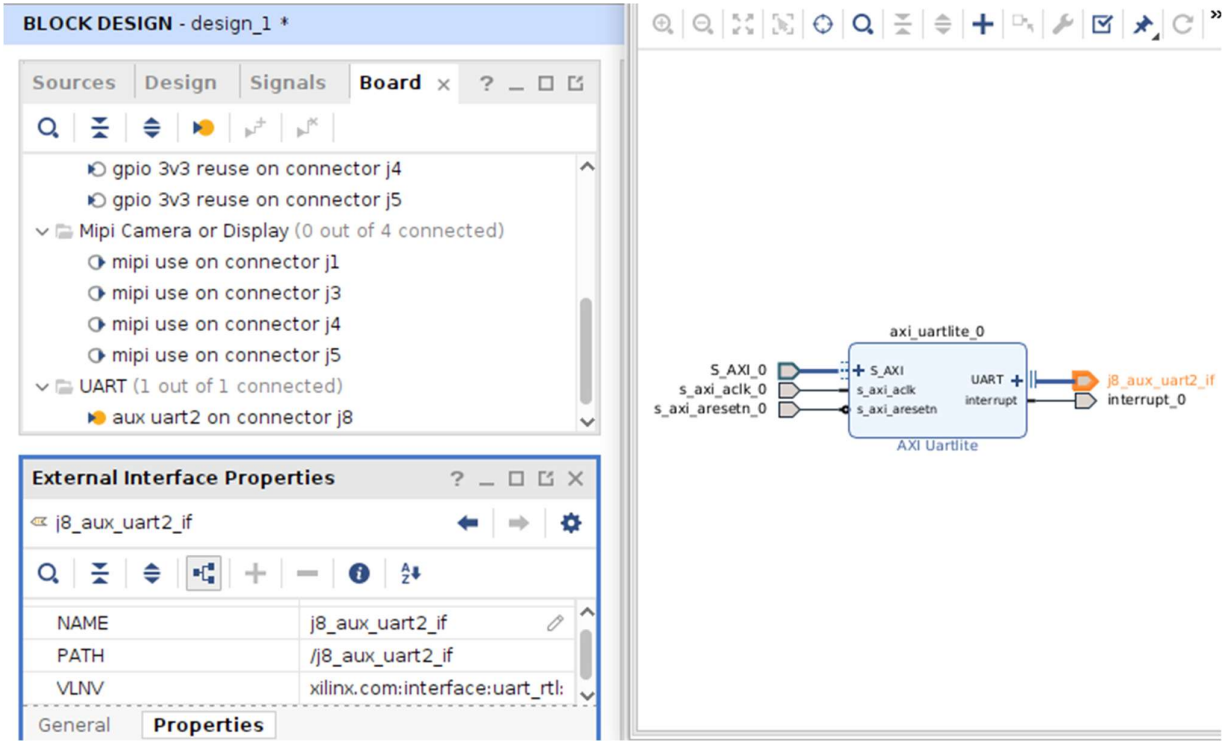


Figure 11 – board file aux uart2 interface

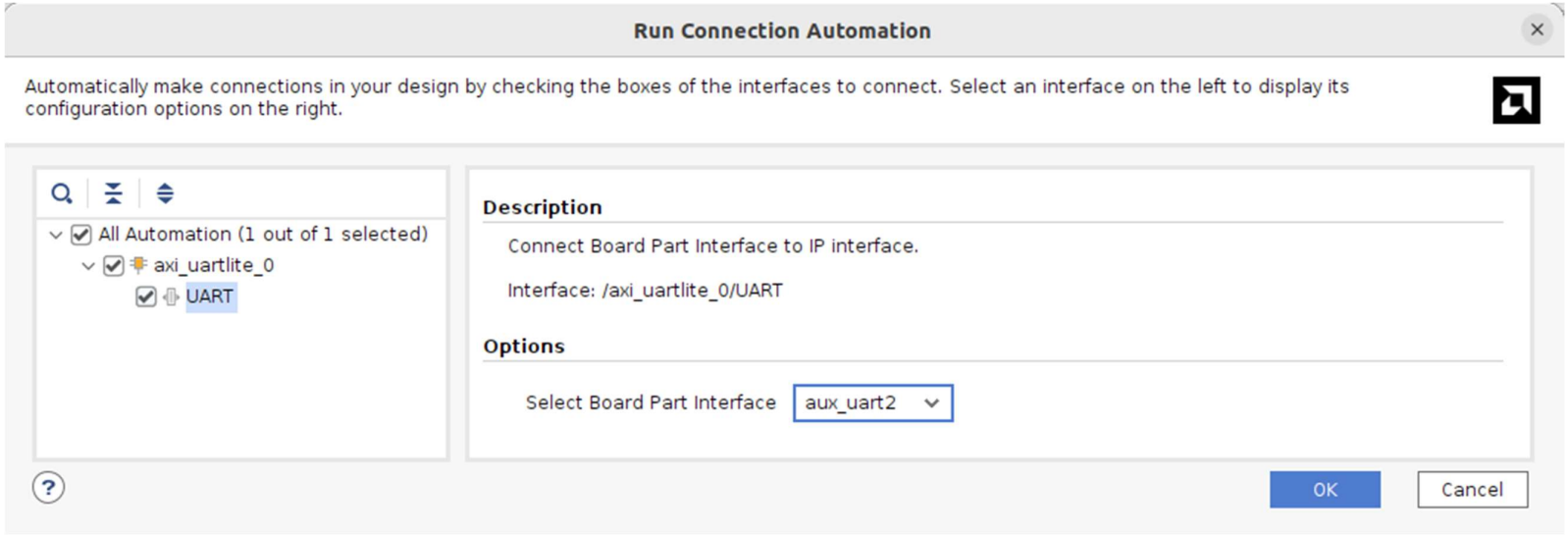


Figure 12 – Connection Automation for aux uart2 interface

3.5.7 j8: gpio

J8

connector pin		board net name	connection type	FPGA pin	note
J8:4		GPIO0	direct connection	B9	
J8:5		GPIO1	direct connection	A11	
J8:6		GPIO2	direct connection	A10	
J8:7		GPIO3	direct connection	A9	

This peripheral is selected when either the “aux gpio on connector j8” is selected from the Board tab or when “AXI GPIO” IP is added to a block design:

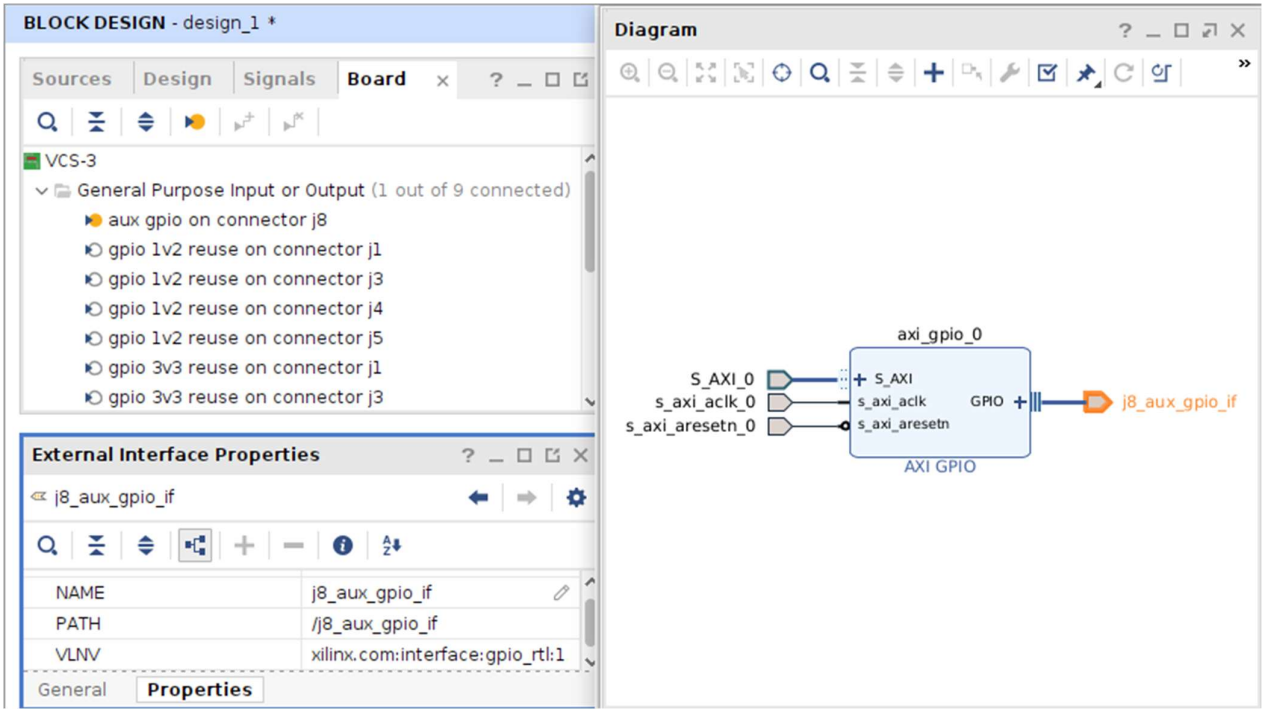


Figure 13 – board file aux gpio interface

3.6 How can I create a simple reference hardware design?

A simple reference design is created here using the board files described above. Open Vivado to create a new project and, when invited to choose a part or board, select the VCS-3.

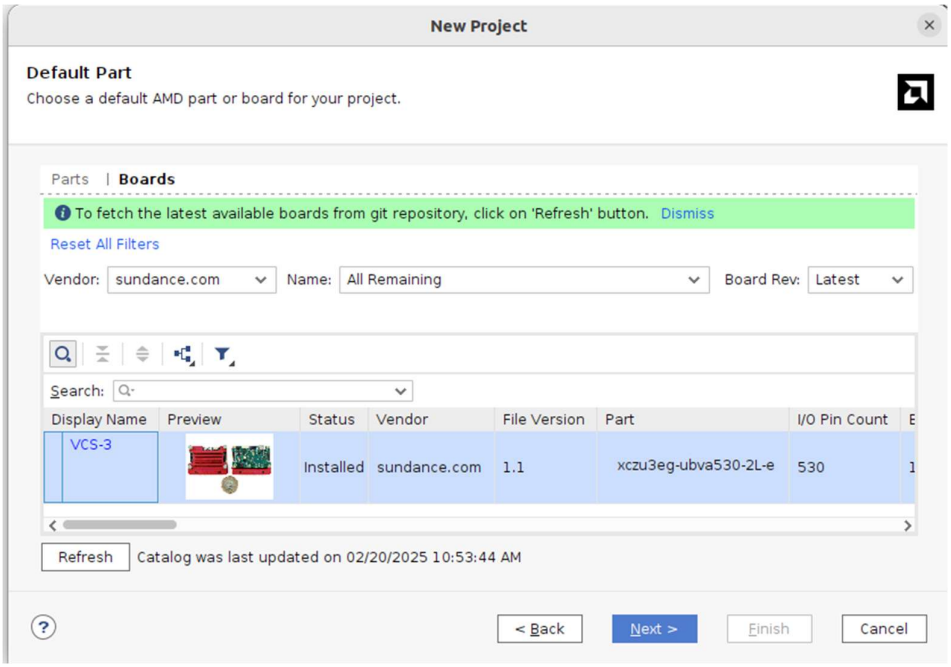


Figure 14 – Selecting VCS3 board file

Once the project is created, click “Create Block Design” and give it a meaningful name.

Notice the Board tab in Block Design window is populated with a range of possible board interfaces.

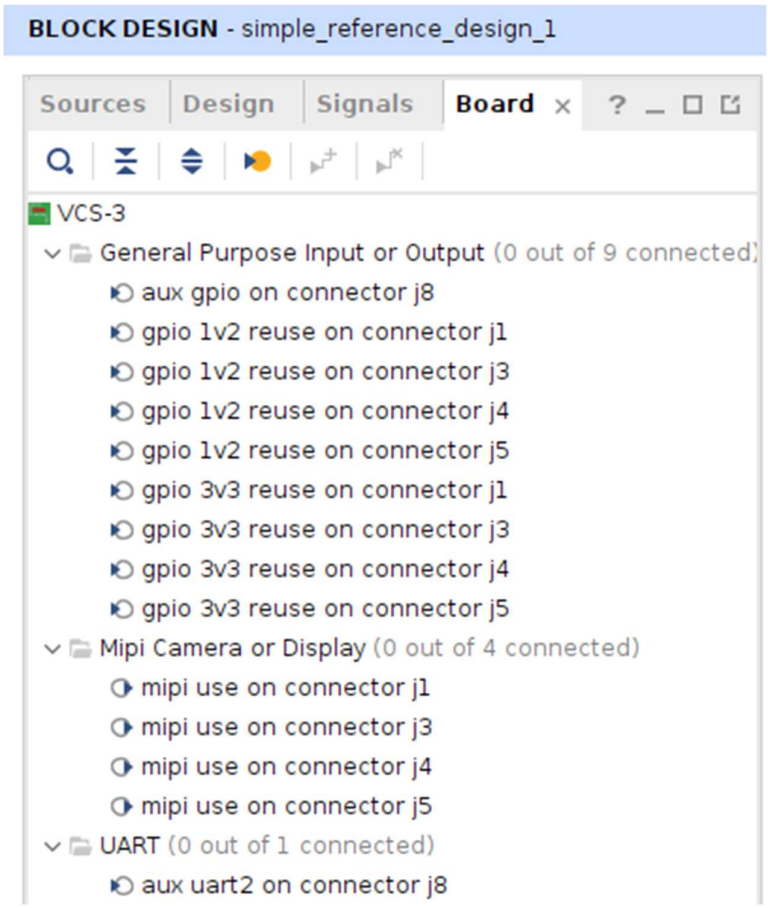


Figure 15 – VCS3 board interfaces in IP Integrator

For this simple design, we'll use all the board connectors for gpio and uarts.

Here j1 is selected and connected to an axi_gpio IP core, the 1v2 logic to the first IP port and 3v3 logic to the second IP port.

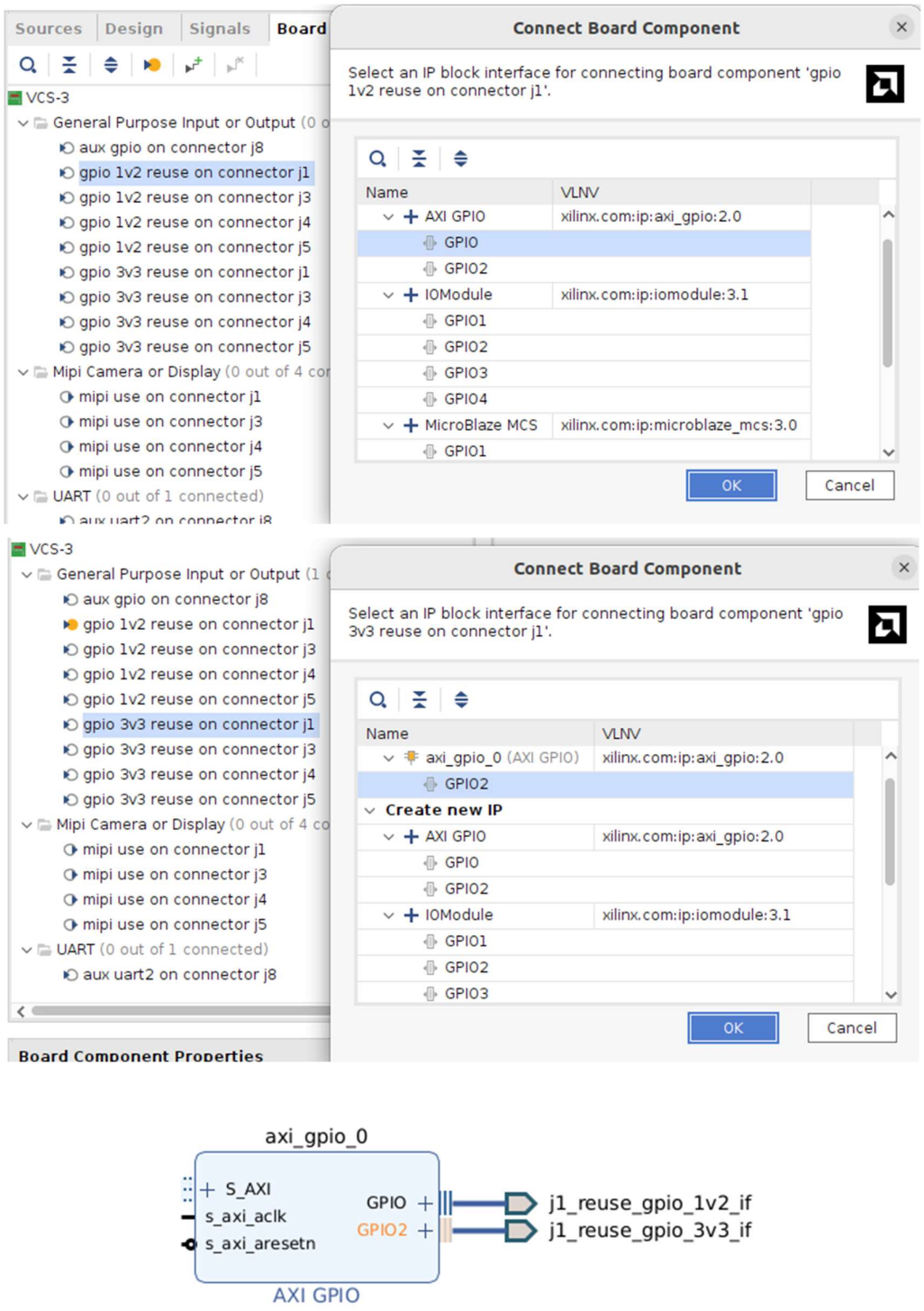


Figure 16 – Selecting axi_gpio for J1 board interface

Doing the same for all the other connectors (but not the mipi ones) …

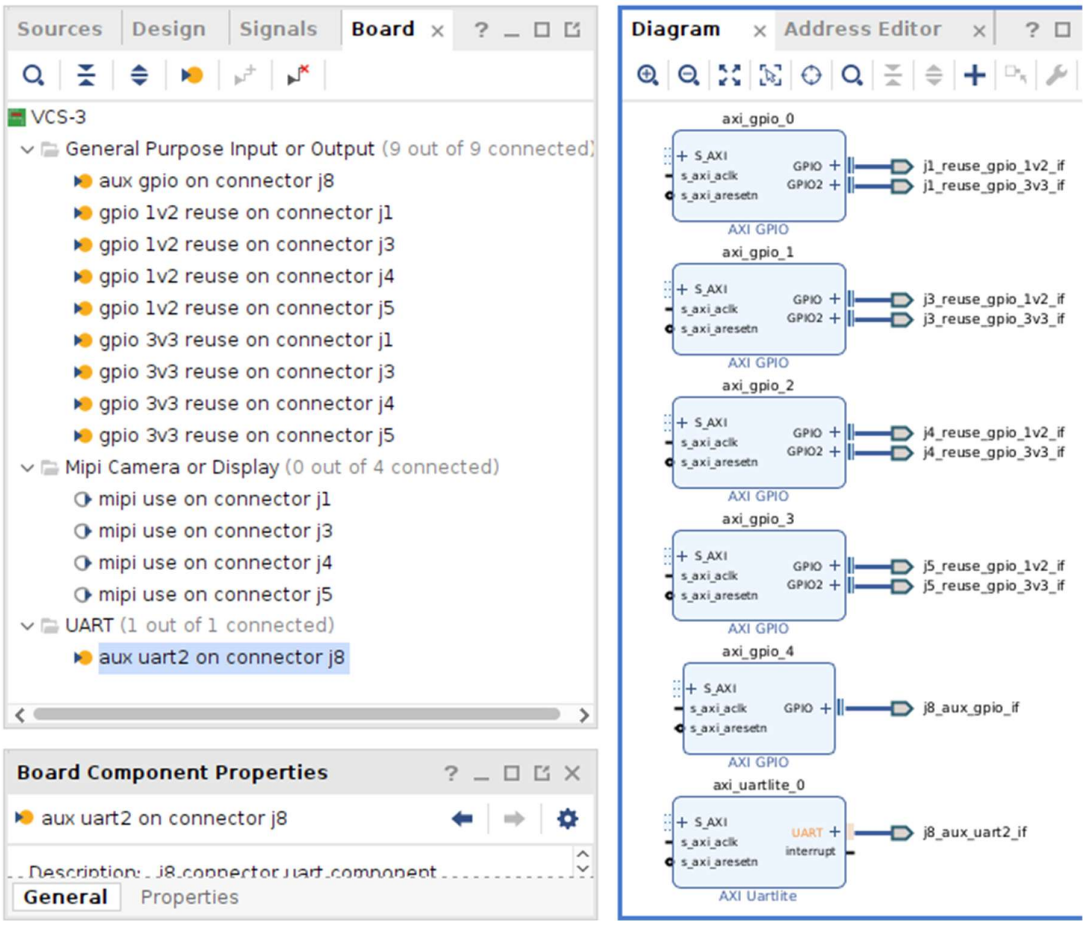


Figure 17 – Complete reuse of all VCS3 board interfaces

After adding a Zynq UltraScale+ MPSoC core to the block design you are invited to use "Designer Assistance": Run Block Automation and Run Connection Automation.

Figure 18 – Adding Zynq block

Block Automation configures the Zynq with board presets and ensures connectivity from the PS side pins to respective peripherals.

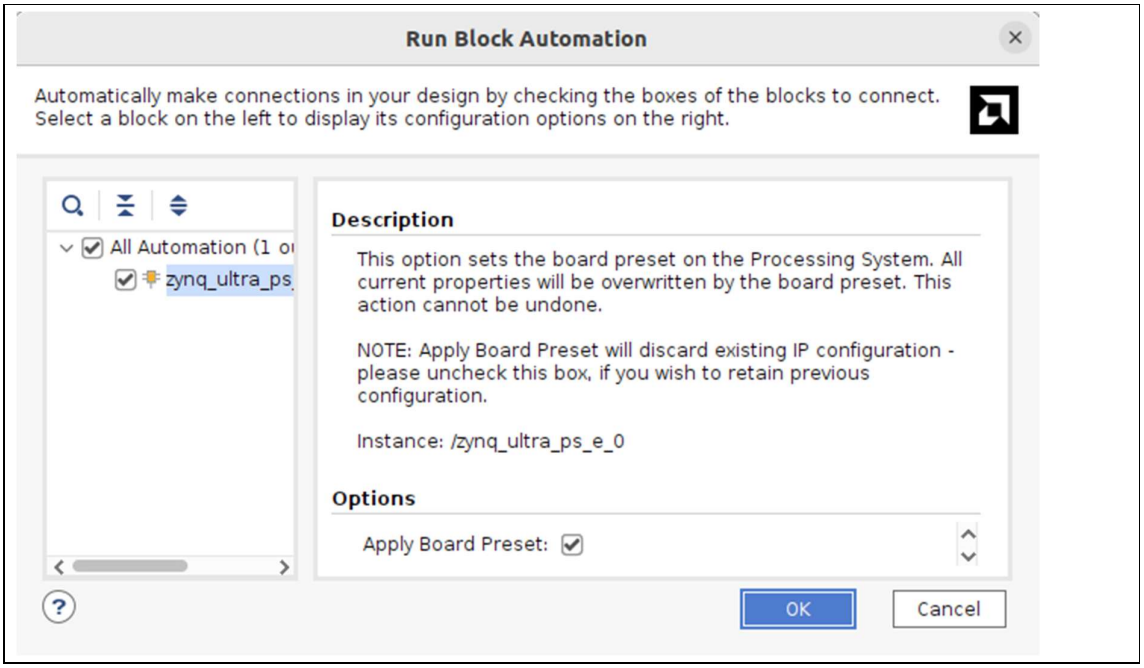


Figure 19 – Running Block Automation for Zynq

Connection Automation wires up all the selected IP cores.

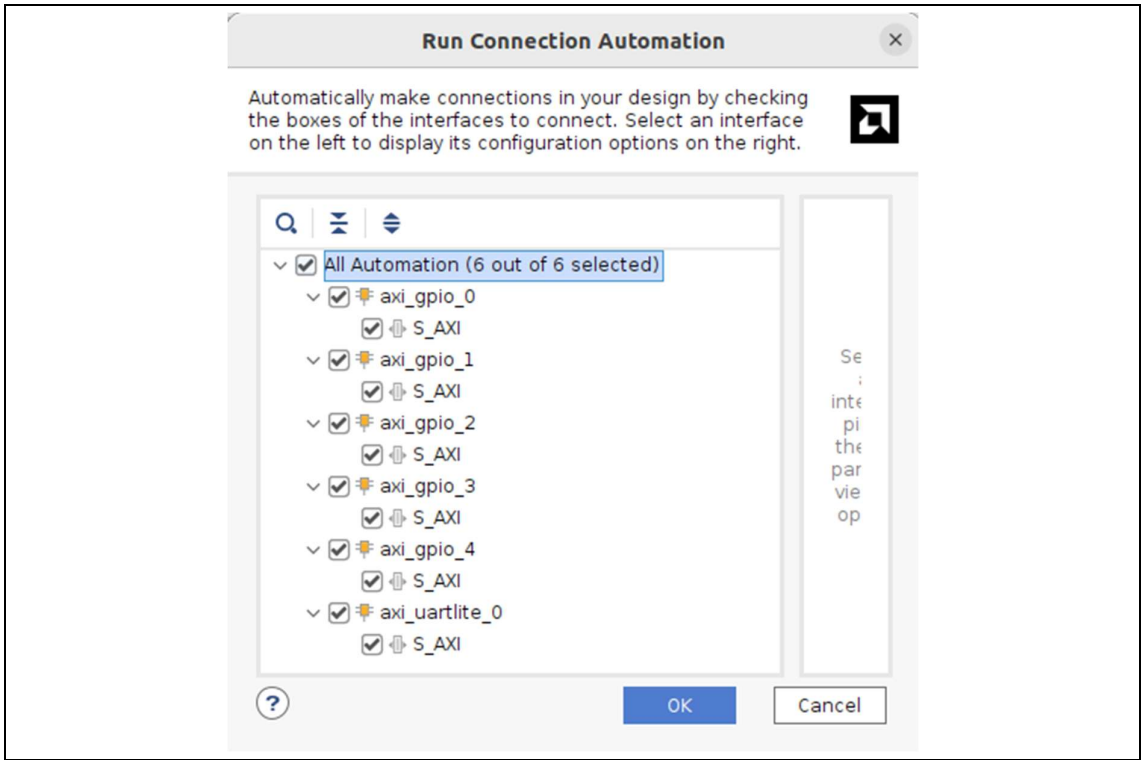


Figure 20 – Running Connection Automation

The resulting block design can be validated using the menu Tools→Validate Design

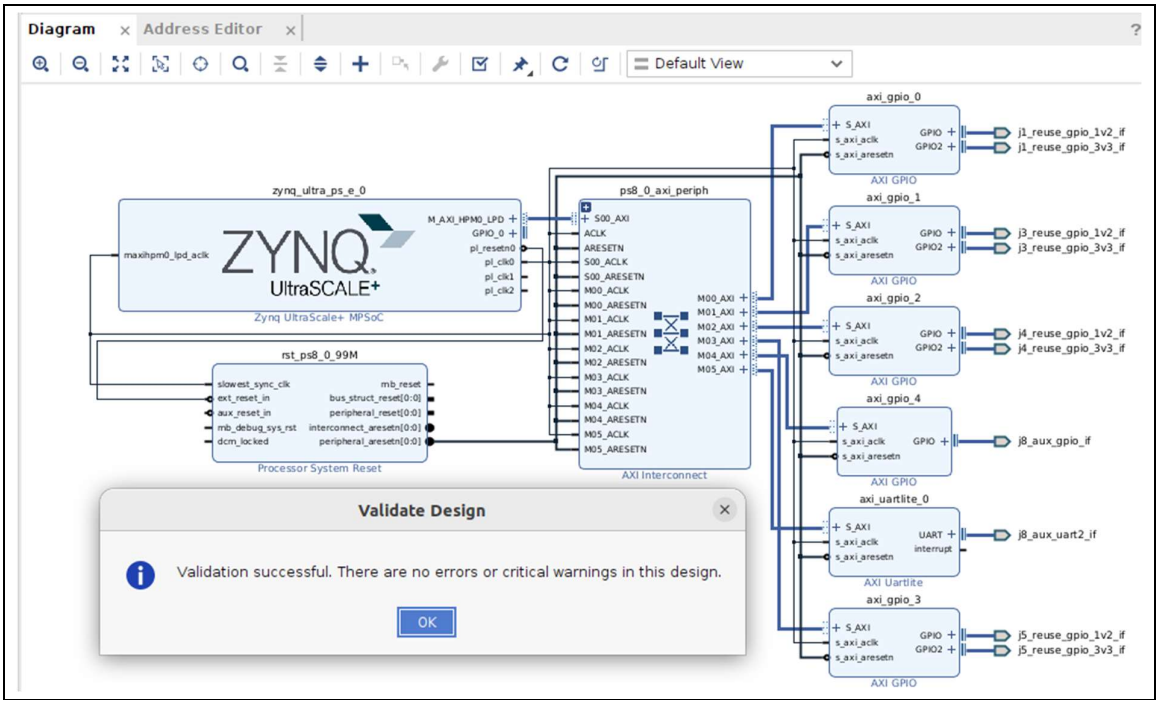


Figure 21 – Validating block design

The block design can now be closed and saved. When it appears in the Project Manager Sources window Hierarchy, you can right-click it to select "Create HDL Wrapper".

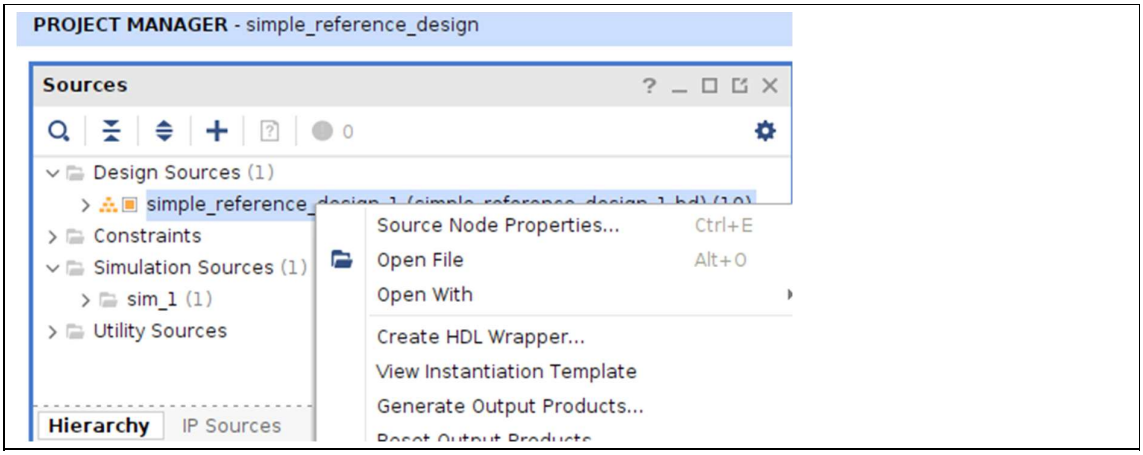
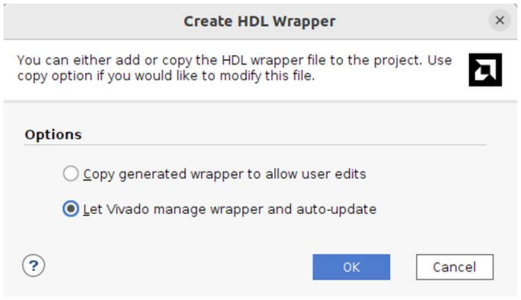


Figure 22 – Creating block design top level HDL wrapper

Choosing to let Vivado manage the wrapper is usually the better option unless you want to add other HDL entities (VHDL) or modules (Verilog) to the top level.

Vivado 2024.2



Vivado 2025.2

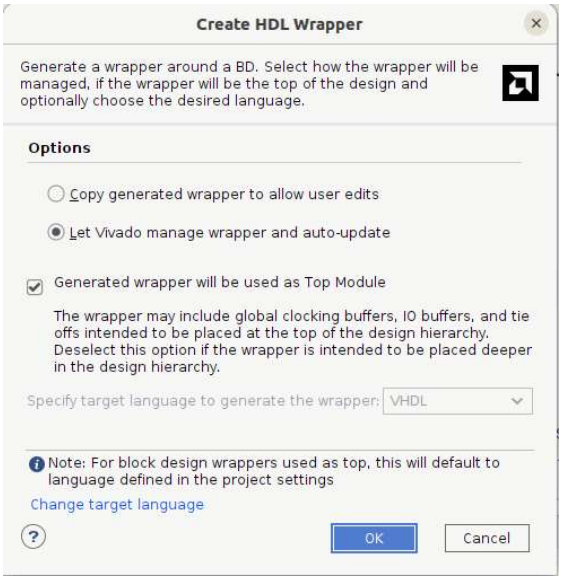


Figure 23 – Letting Vivado manage top level HDL wrapper

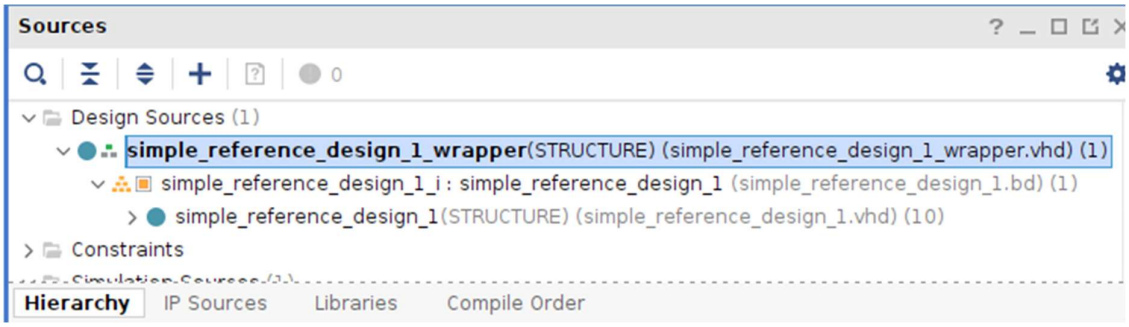


Figure 24 – Top level wrapper appearing in sources hierarchy

The generated wrapper here is in VHDL but it could be in Verilog if the project default HDL had been set to that.

To generate the complete hardware project, just select “Generate Bitstream” in the Project Manager window and choose Yes or OK to any popup dialog boxes. This process will:

- Generate output products for each IP core in the block diagram
- Run synthesis for each of them and also the top level HDL wrapper
- Run Implementation (place and route)
- Produce a bitstream that can be downloaded to the VCS3

The whole process may take some time … relax! You’ll know when it’s finished when you see this in the top right corner of Vivado:

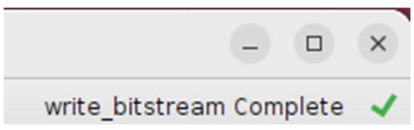
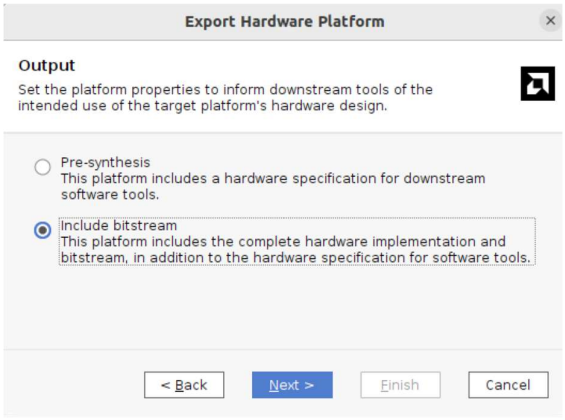


Figure 25 – Bitstream complete

This hardware design on its own does nothing! It needs a software application running on the PS (ARM processor) to drive it. So, the last step here is to export the hardware design so it can be imported into Vitis.

To export the hardware, use the menu File→Export→Export Hardware (NB: In 2025.2, I found that after ticking the "Include binary", the expected .xsa file didn't show where expected; very strange!)

Vivado 2024.2



Vivado 2025.2

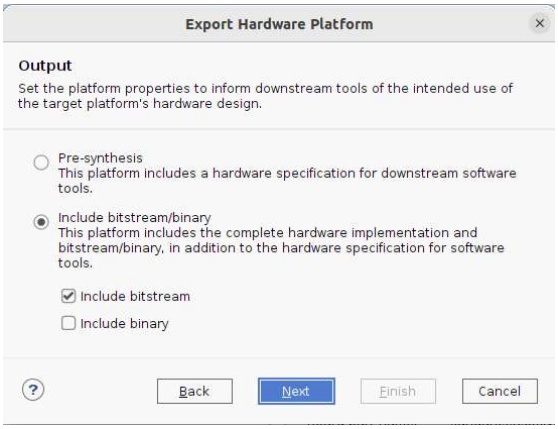


Figure 26 - Exporting hardware including the bitstream

Continuing with the above simple reference design …

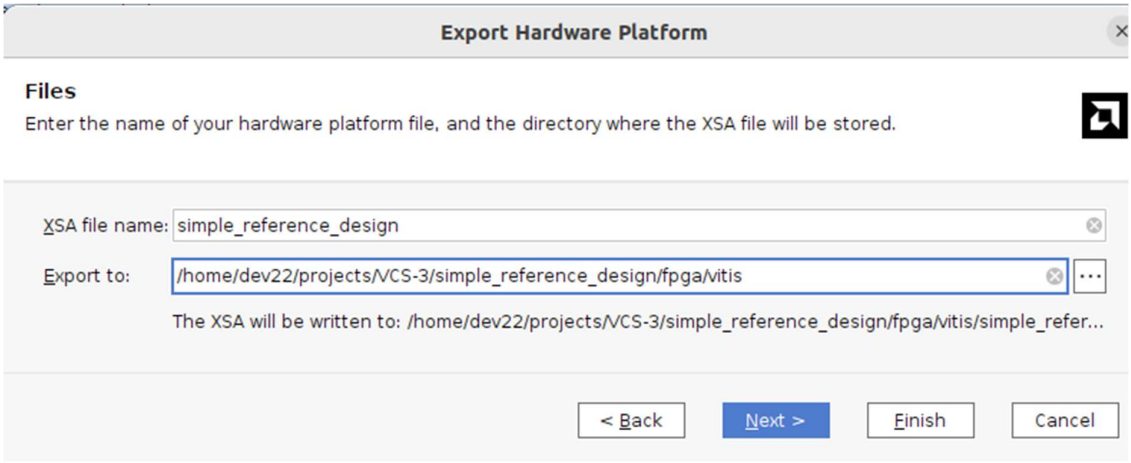


Figure 27 – Exporting hardware to a directory where Vitis will use it

3.7 How can I archive and reproduce my simple reference design?

Vivado makes it easy for you to archive your block design as a tcl scripts (tcl stands for Tool Command Language). You just need to create the script using the menu File→Export→Export block Design.

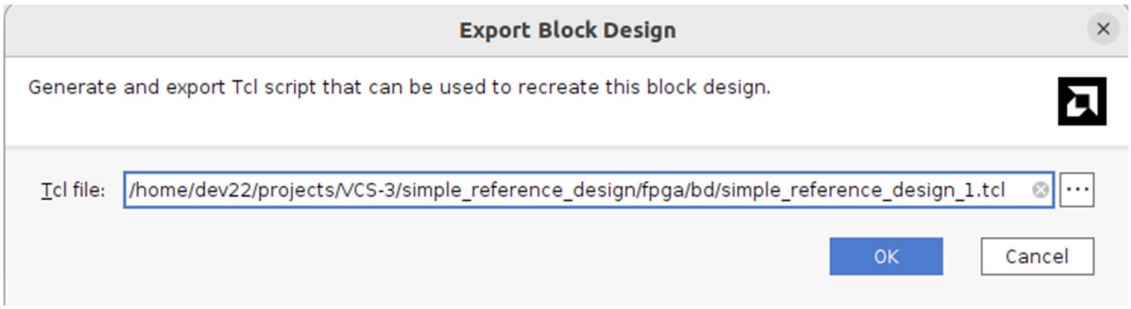


Figure 28 – Exporting block design as a tcl script

To recreate the block design, just open a new project for VCCS-3 and run the script.

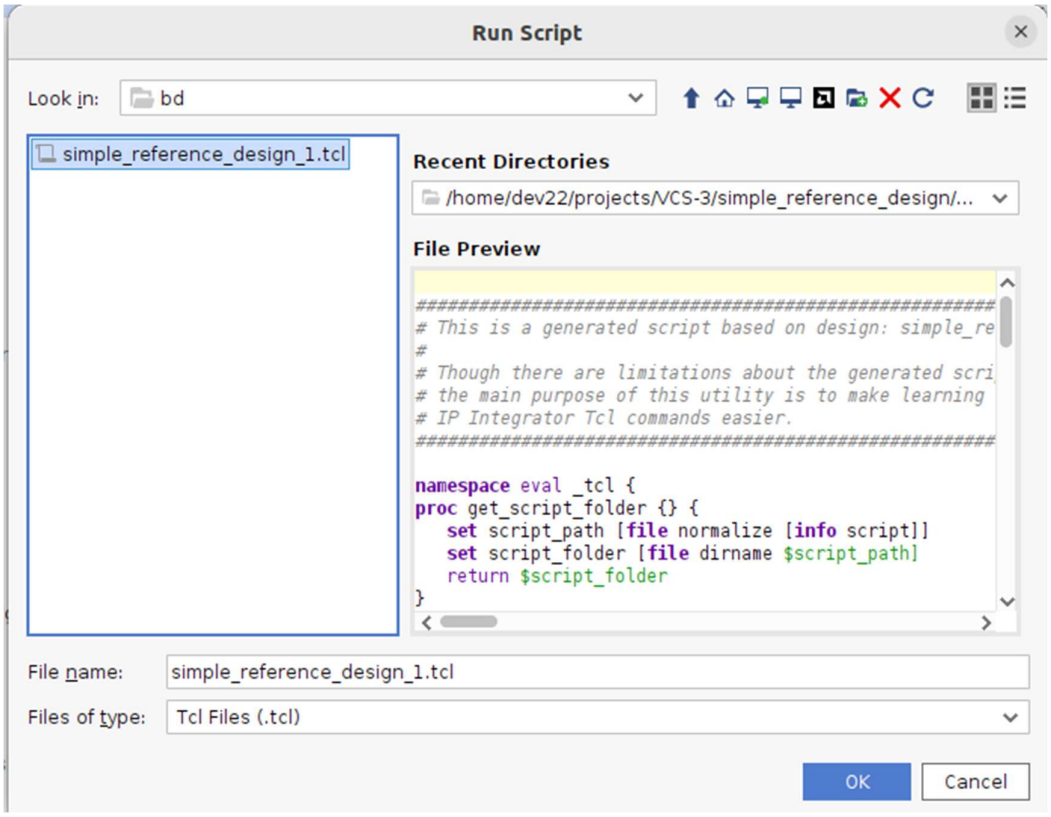


Figure 29 – Recreating block design using tcl script

You can then recreate the top level wrapper and generate the bitstream as before. The above tcl script is saved in the VCS-3 archive for you to try.

4 Software

Before any software can be written for the VCS3 processor side (PS) the hardware developed previously must be exported from Vivado and imported into Vitis.

4.1 Creating Vitis platform component from Vivado hardware

When starting Vitis you must choose a working directory. Then, to import the above hardware .xsa file, use the menu File→New Component→Platform and follow the sequence of dialogs.

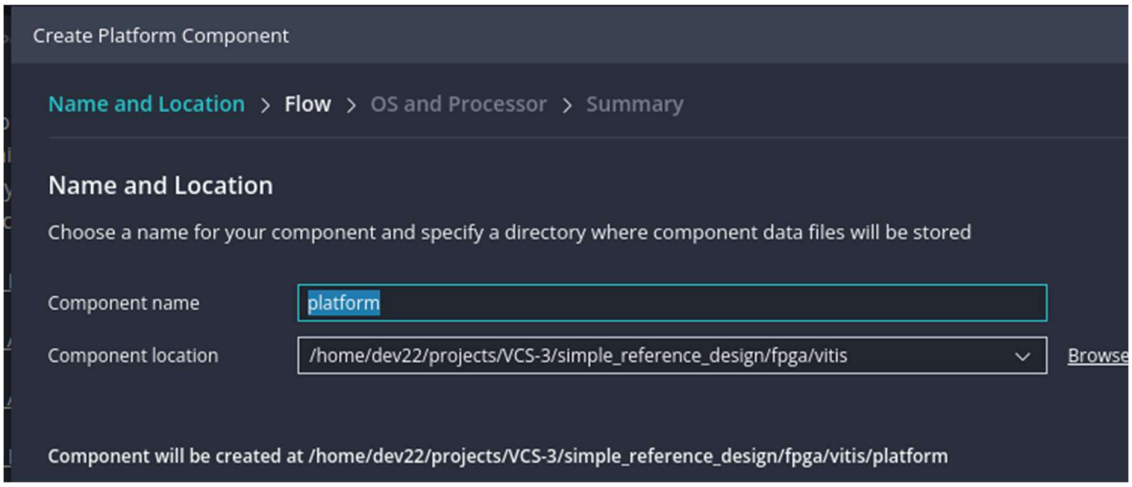


Figure 30 – Creating platform in Vitis – naming platform component

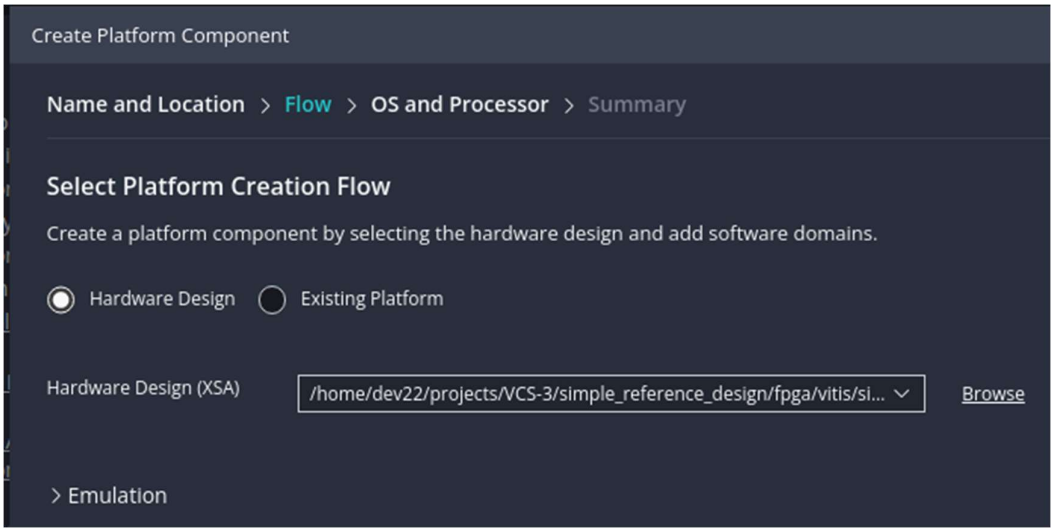


Figure 31 – Creating platform in Vitis – finding .xsa file

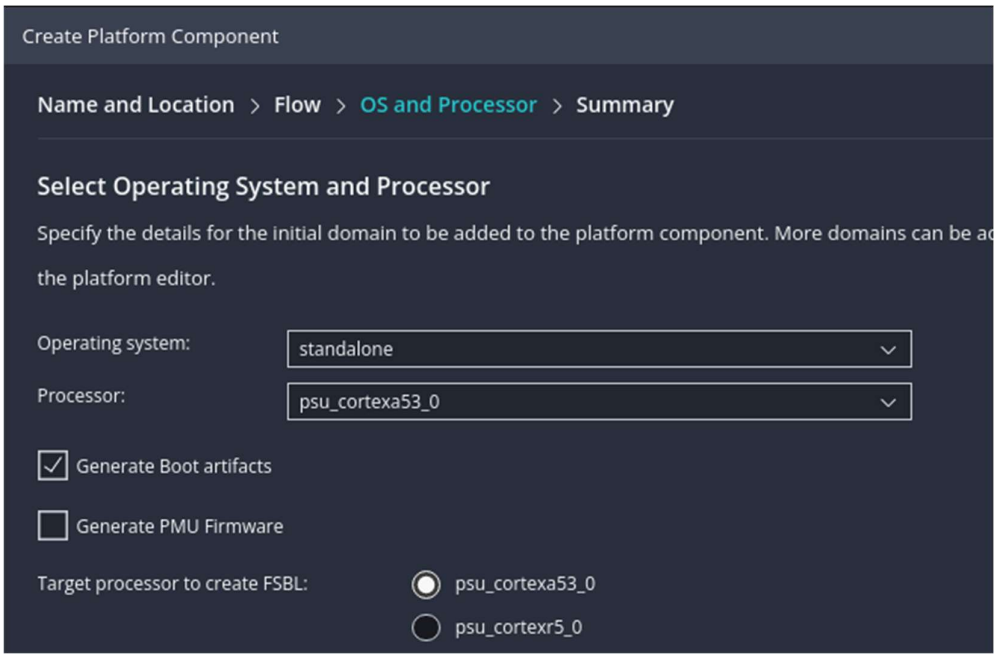


Figure 32 – Creating platform in Vitis – selecting processor and operating system

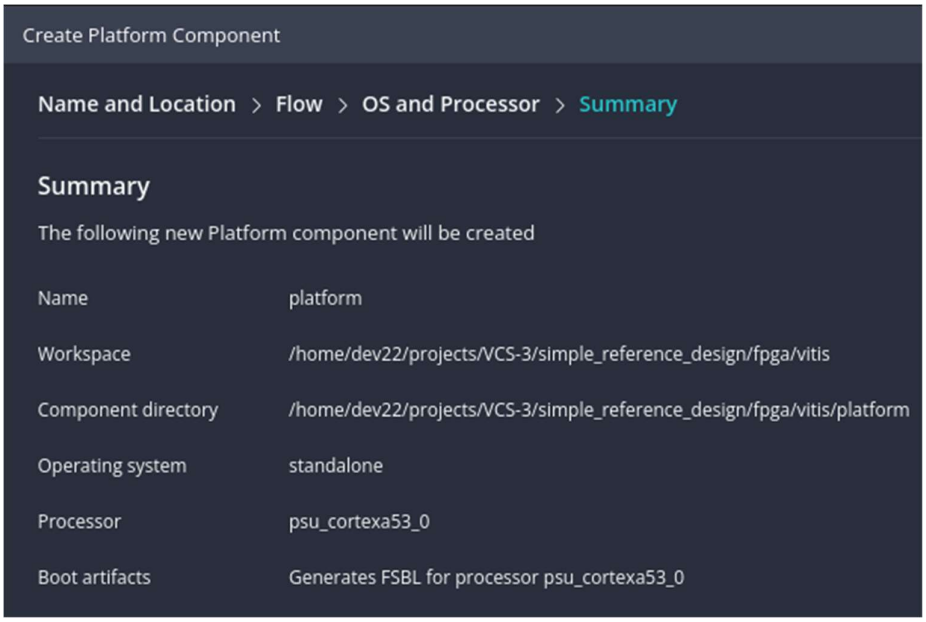


Figure 33 – Creating platform in Vitis – summary

This platform component can now be built by clicking the Build icon

Figure 34 – Building platform in Vitis

4.2 How can I create a “hello world” app?

There are a few options for creating a new software application. One of the easiest is to copy an example application and elaborate it to your own specification. To do this, click on the examples icon and select one of the many preprepared applications e.g. “Hello World” and follow the sequence of dialogs.

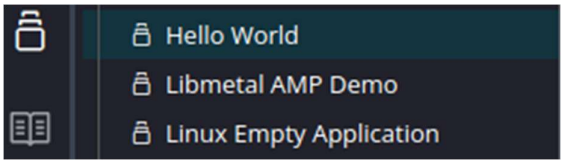


Figure 35 – Selecting example application

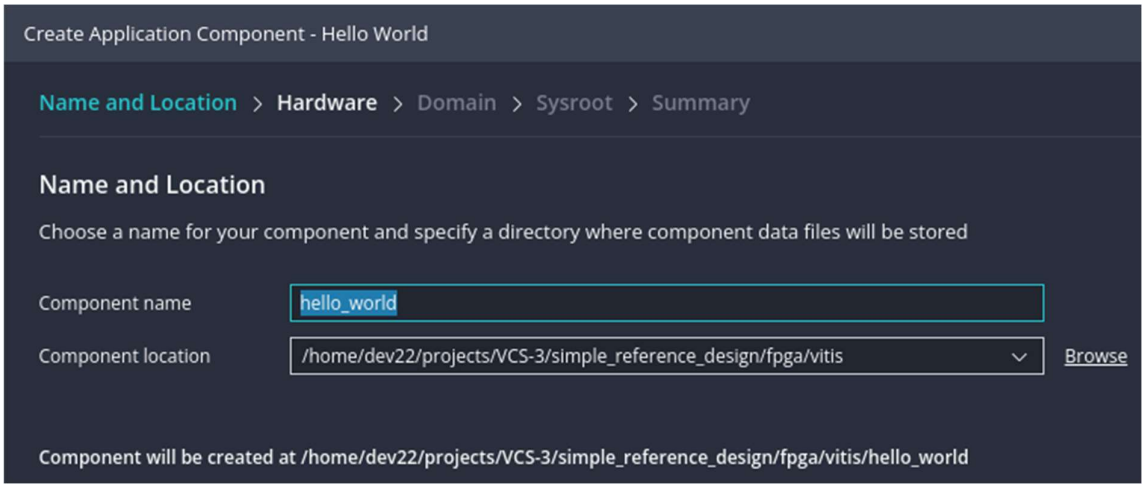


Figure 36 – Creating an application in Vitis – choosing application component name

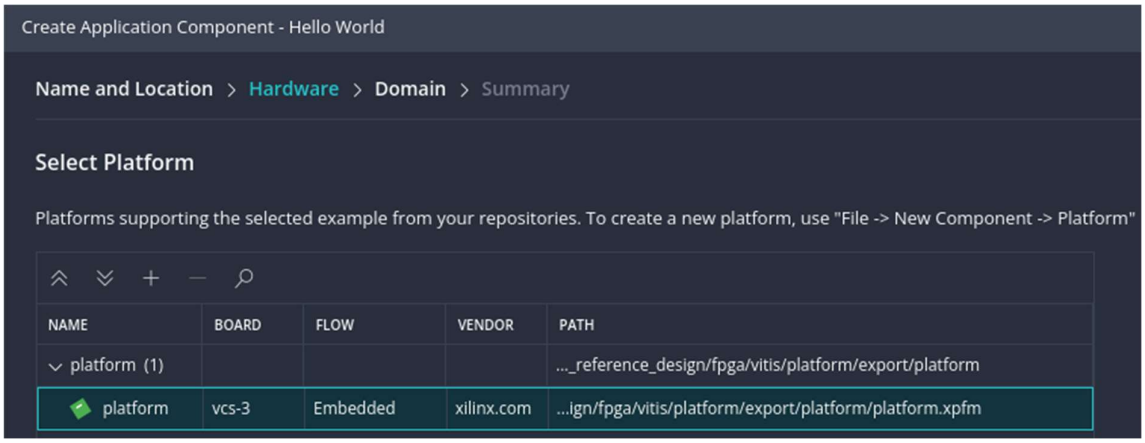


Figure 37 – Creating an application in Vitis – associating hardware platform

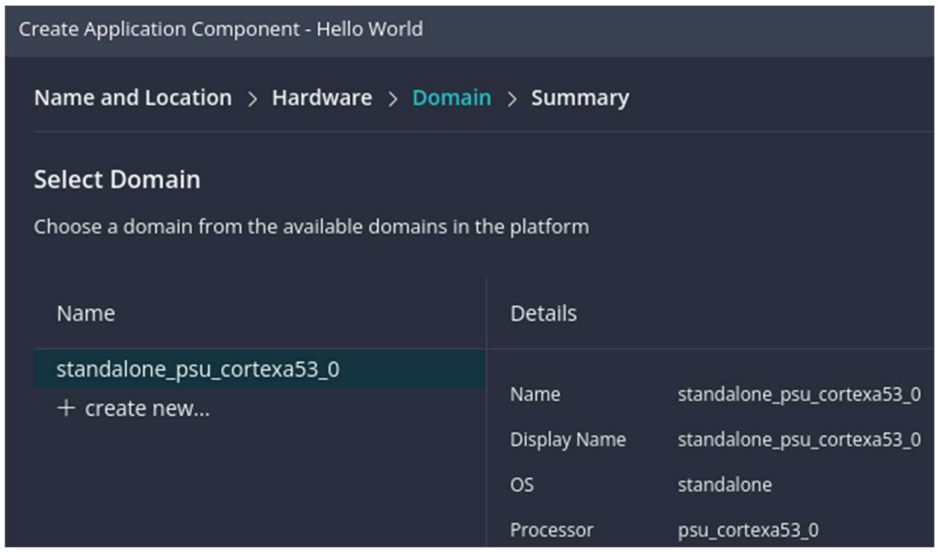


Figure 38 – Creating an application in Vitis – choosing processor domain

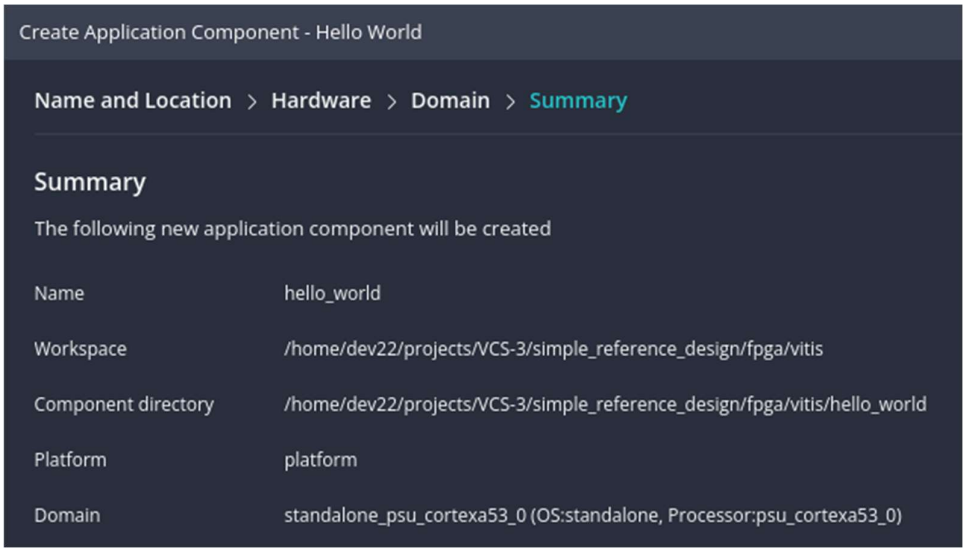


Figure 39 – Creating an application in Vitis – summary

This application component can now be built by clicking the Build icon.

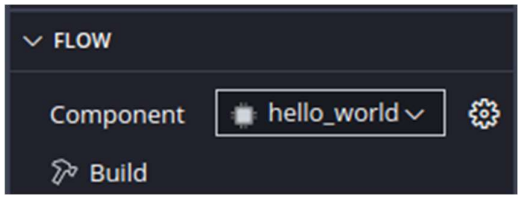


Figure 40 – Building application in Vitis

Before proceeding, choose if you want to rebuild the application component if the platform changes (this option is possibly new to 2025.2):

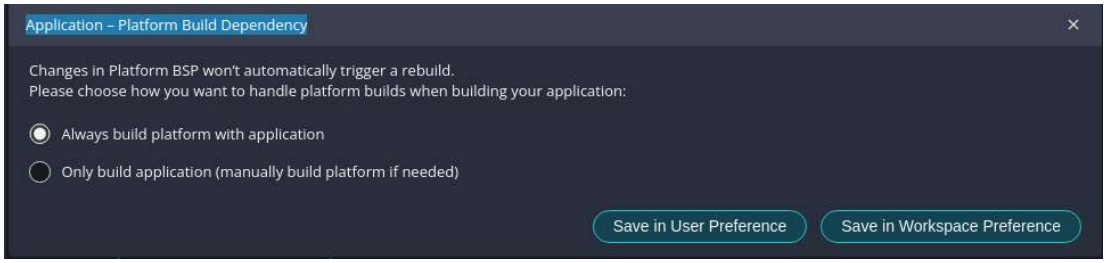


Figure 41 – Building application in Vitis

4.3 Running "hello world" app on hardware?

The built application can now be downloaded and run on the VCS3 hardware depending on your JTAG connection. If you have the VCS3 development kit (as described in "VCS3 Development Kit Getting Started Guide") then the easiest way is to invoke the Lynsyn Xilinx Virtual Cable driver as shown here.

NOTE: If you are not using the VCS3 Development Kit (i.e. not using the Lynsyn), skip to debug session.

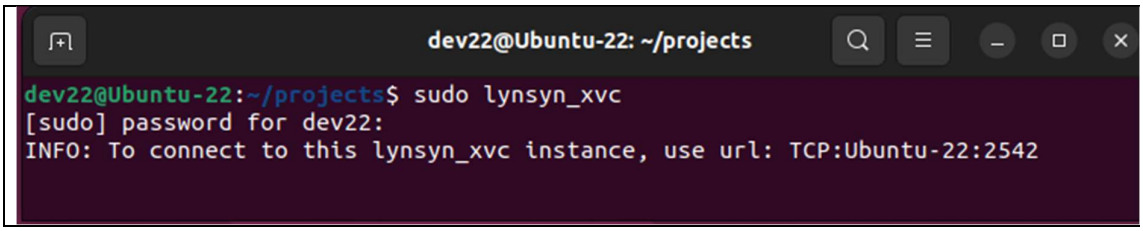


Figure 42 – Running Lynsyn Xilinx Virtual Cable

If running Vitis from a VM don't forget to claim the required USB devices from the host machine..

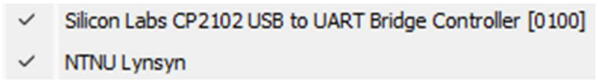


Figure 43 – VM taking control of required host USB device

So that Vitis can use the Lynsyn, open a terminal using menu Terminal→New Terminal and invoke xsct and run command line: connect -xvc-url localhost:2542

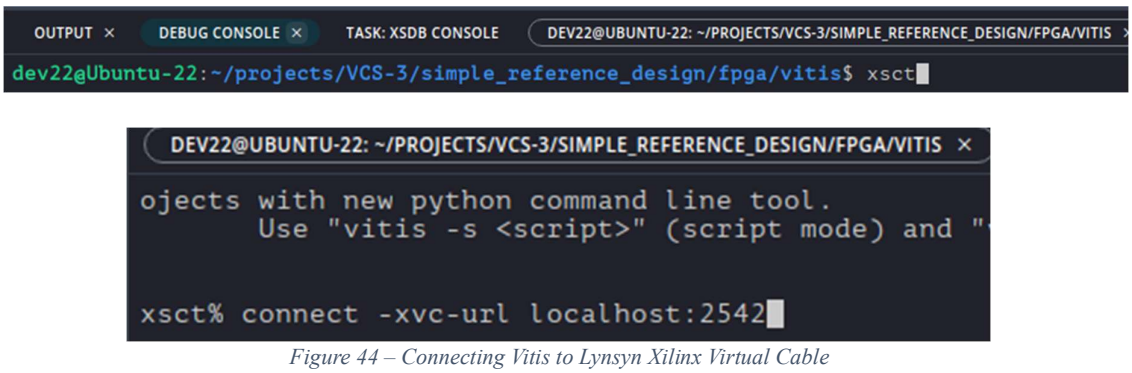


Figure 44 – Connecting Vitis to Lynsyn Xilinx Virtual Cable

The debug session can now be started …

```
15:36:04 INFO  : XSDB server has started successfully from frontend.
15:36:05 INFO  : Connection to XSDB Server established.
15:36:05 INFO  : connect -url tcp:127.0.0.1:3121
15:36:05 INFO  : Done
15:36:13 INFO  : bpremove -all
15:36:13 INFO  : Context for 'APU' is selected.
15:36:14 INFO  : System reset is completed.
15:36:17 INFO  : 'after 3000' command is executed.
15:36:17 INFO  : 'targets -set -filter {jtag_cable_name =~ "Xilinx Virtual Cable
localhost:2542" && level==0 && jtag_device_ctx=="jsn-XVC-localhost:2542-14710093-0"}' command
is executed.
15:36:53 INFO  : Device configured successfully with "/home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/hello_world/_ide/bitstream/simple_reference_design.bit"
15:36:53 INFO  : loadhw -hw /home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/platform/export/platform/hw/simple_reference_design.xsa
-mem-ranges [list {0x80000000 0xbfffffff} {0x400000000 0x5fffffff} {0x1000000000
0x7fffffffff}]
15:36:53 INFO  : targets -set -nocase -filter {name =~"APU*"}
15:36:53 INFO  : source /home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/hello_world/_ide/psinit/psu_init.tcl
15:37:06 INFO  : psu_init
15:37:07 INFO  : after 1000
15:37:08 INFO  : psu_ps_pl_isolation_removal
15:37:09 INFO  : after 1000
15:37:09 INFO  : psu_ps_pl_reset_config
15:37:09 INFO  : targets -set -nocase -filter {name =~ "*A53*#0"}
15:37:11 INFO  : rst -processor
15:37:15 INFO  : dow /home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/hello_world/build/hello_world.elf
15:37:15 INFO  : bpadd main
15:37:16 INFO  : con
15:37:17 INFO  : Testing the connection for 127.0.0.1
```

… and halts at the source code entry point:

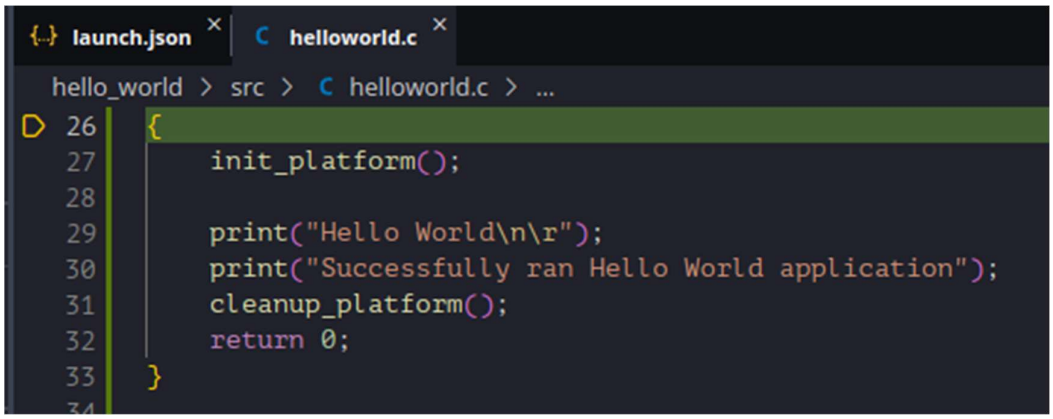


Figure 45 – Vitis debug session for helloworld

If a terminal emulator is attached to the Lynsyn USB serial device, the message can be seen from the application when run

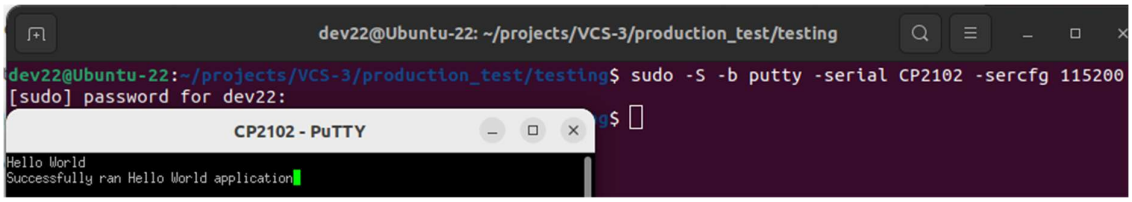


Figure 46 – Terminal emulator showing output of helloworld

[see *section 5.2* for help on debugging]

4.4 How can I create a boot image for “hello world”?

Simply by selecting “Create Boot Image” in the Flow window …

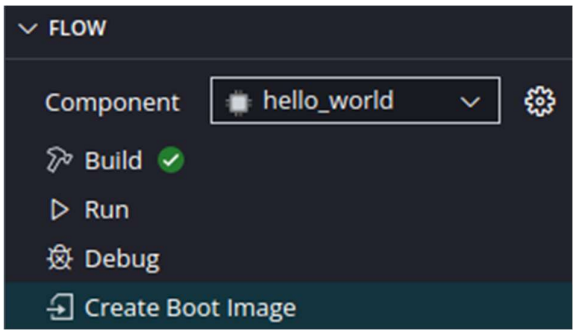


Figure 47 – Running Create Boot Image in Vitis

… and following dialog. Here, we don’t bother about encryption, so most of the fields are empty.

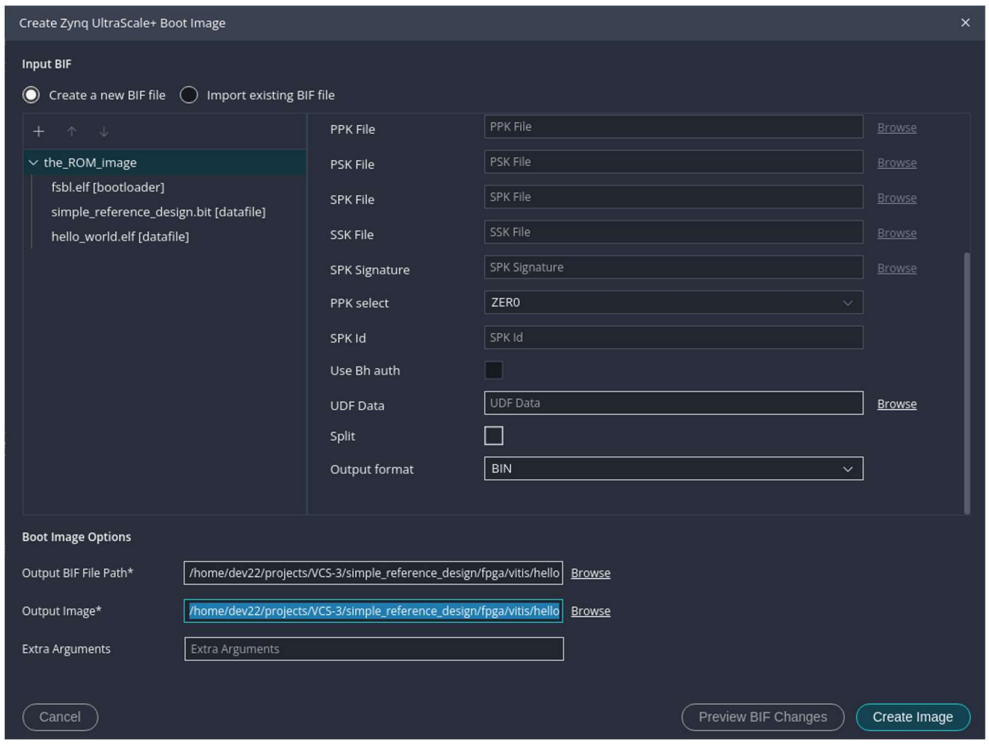


Figure 48 – Create Boot Image in Vitis

The result of selecting “Create Image” is a .bin file called “BOOT.bin” and a .bif file, in this case, called “hello_world.bif”

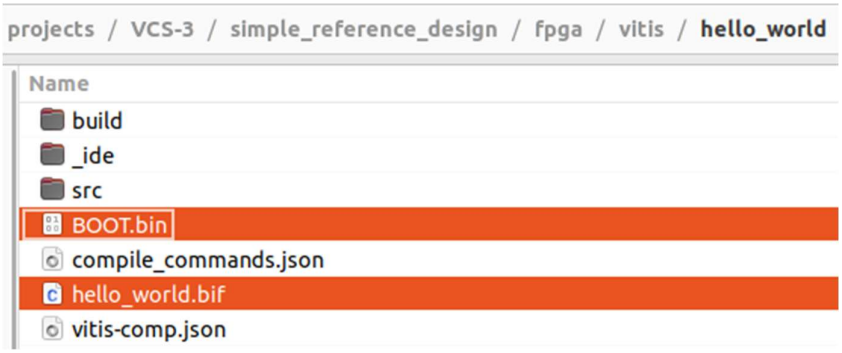


Figure 49 – Output of Create Boot Image

The bif file is simply a recipe for creating the bin file from three images:

- first stage boot loader (FSBL) .elf file
- programmable logic (PL) firmware application .bit file
- processor side (PS) software application .elf file

```
//arch = zynqmp; split = false; format = BIN
the_ROM_image:
{
 [bootloader, destination_cpu = a53-0]/home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/platform/export/platform/sw/boot/fsbl.elf
 [destination_device = pl]/home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/hello_world/_ide/bitstream/simple_reference_design.bit
 [destination_cpu = a53-0, exception_level = el-3]/home/dev22/projects/VCS-
3/simple_reference_design/fpga/vitis/hello_world/build/hello_world.elf
}
```

The .bif file can be reused to recreate the .bin file when any image is changed.

The .bin file can be programmed into eMMC or SPI ROM memory on the VCS3 using the menu Vitis→Program Flash.

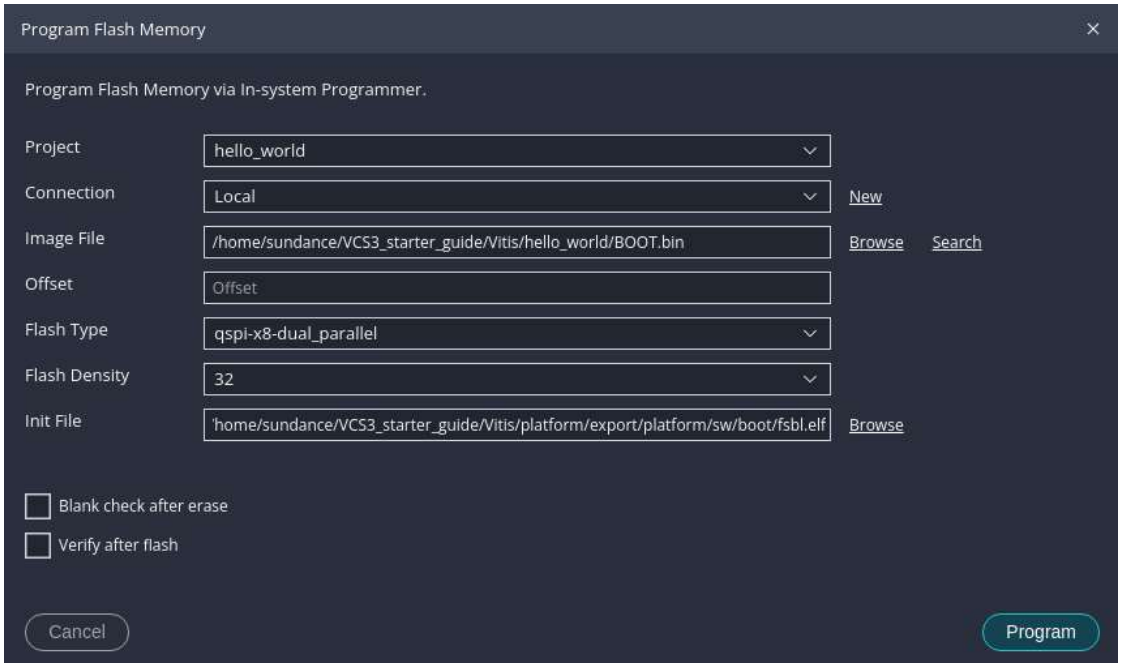


Figure 50 – Programming flash in Vitis

The process of erasing flash, programming verifying may take some time … but you can monitor progress in the Output window:

```
…
…
device 0 offset 0x578000, size 0xfd4
SF: 4052 bytes @ 0x578000 Read: OK
ZynqMP> cmp.b FFFC0000 FFFC8000 FD4

Total of 4052 byte(s) were the same
ZynqMP> INFO: [Xicom 50-44] Elapsed time = 836 sec.
Verify Operation successful.

Flash Operation Successful

Program flash finished.
----------------------------------------------------------------------------
```

```
[04/03/2025, 16:21:29]: Program flash with id 'f3edcb0b-6395-42d3-9101-d47d5cd22f3c' ended.
```

4.5 Selecting boot memory

By setting dual switch SW1 on the VCS3 as described below, the Zynq device will be configured from the selected memory when powering up the board.

Switch 1	Switch 2	
OFF	OFF	JTAG
ON	OFF	eMMC
OFF	ON	QSPI ROM
ON	ON	

Figure 51 – Setting boot mode

5 Appendix

Additional information to catch and help solve potential issues.

5.1 Cloning the GitHub repository

Cloning the https://github.com/SundanceMultiprocessorTechnology/VCS-3.git repository may be difficult because of access permissions.

- In terminal:
  - git init
  - git config --global user.name <GitHub account username>
  - git config --global user.email <GitHub account email>
  - ssh-keygen -t ed25519 -C your_email@example.com
  - eval "$(ssh-agent -s)"
  - ssh-add ~/.ssh/id_ed25519
  - cat ~/.ssh/id_ed25519.pub
- Go to: https://github.com/settings/keys
  - Click **"New SSH key"**
  - Paste your key and give it a name (e.g., "My Linux Laptop")
- Back in the terminal, clone repository using:
  - git clone git@github.com:SundanceMultiprocessorTechnology/VCS-3.git

Figure 52 – Creating an SSH key

5.2 Application debug help

- Flow > Debug > Open Settings
  - Check that 'FSBL' is selected next to 'Board Initialisation'
- Click 'Debug' in the Flow window.
- Once the debug program has stopped at the source code entry point, click 'Continue' or press F5 to complete the debug.
- Observe the 'Hello world' message in your PuTTy (or alternate output if not using PuTTy) terminal!

Figure 53 – Application debug help

If FSBL Board Initialisation mode fails, fully power cycle the board and retry.